



Club deals in leveraged buyouts[☆]

Micah S. Officer^{a,*}, Oguzhan Ozbas^b, Berk A. Sensoy^c

^a College of Business Administration, Loyola Marymount University, Los Angeles, CA 90045, USA

^b Marshall School of Business, University of Southern California, Los Angeles, CA 90089, USA

^c Fisher College of Business, The Ohio State University, Columbus, OH 43210, USA

ARTICLE INFO

Article history:

Received 29 October 2008

Received in revised form

26 August 2009

Accepted 8 December 2009

Available online 1 June 2010

JEL classification:

G34

G38

K21

Keywords:

Club deals

Leveraged buyouts

Private equity firms

ABSTRACT

We analyze the pricing and characteristics of club deal leveraged buyouts (LBOs)—those in which two or more private equity partnerships jointly conduct an LBO. Using a comprehensive sample of completed LBOs of U.S. publicly traded targets conducted by prominent private equity firms, we find that target shareholders receive approximately 10% less of pre-bid firm equity value, or roughly 40% lower premiums, in club deals compared to sole-sponsored LBOs. This result is concentrated before 2006 and in target firms with low institutional ownership. These results are robust to controls for target and deal characteristics, including size, Q, measures of risk, and time and industry fixed effects. We find little support for benign motivations for club deals based on capital constraints, diversification motives, or the ability of clubs to obtain favorable debt amounts or prices, but it is possible that the lower pricing of club deals is an inadvertent byproduct of an unobserved benign motivation for club formation.

© 2010 Elsevier B.V. All rights reserved.

1. Introduction

Following the leveraged buyout (LBO) boom of the 1980s and the relatively quiet 1990s, LBO activity again boomed in the 2001–2007 period before collapsing in 2008. Total LBO deal volume in the United States alone

rose from approximately \$30 billion in 2001 to over \$450 billion in 2007 (Thomson Financial M&A Review, 2007). Kaplan and Strömberg (2009) report that total equity capital commitments to U.S. private equity funds reached \$228.0 billion in 2007, or 1.57% of total U.S. stock market capitalization.¹

This sharp increase in private equity activity has reignited the debate among economists and policymakers regarding the efficiency and welfare implications of LBOs. Proponents of the private equity/LBO model stress the potential for creating value in target firms through operating improvements, reduced agency costs of free

[☆] We thank Ola Bengtsson, Harry DeAngelo, Stu Gillan, Mike Hertzel, Yael Hochberg, Alexander Ljungqvist, Rich Mathews, Clara Raposo, Mike Stegemoller, Per Strömberg, Mark Westerfield, two anonymous referees, seminar participants at Chapman University, Cornerstone Research, CRA International, Duke University, Fordham University, ISCTE Business School (Lisbon), Loyola Marymount University, Ohio State University, the Universities of California Riverside, North Carolina, Rochester, and Southern California, Vanderbilt University, and Washington University in St Louis, and conference participants at the 2009 EFA meetings and the 2008 Frontiers of Finance conference for helpful comments, and Derek Horstmeyer for able research assistance. This research began while Officer and Sensoy were on the faculty at the Marshall School of Business (USC).

* Corresponding author.

E-mail addresses: micah.officer@lmu.edu (M.S. Officer), ozbas@usc.edu (O. Ozbas), bsensoy@fisher.osu.edu (B.A. Sensoy).

¹ We use the terms LBO and private equity sponsored deal interchangeably, as do Kaplan and Strömberg (2009). Some authors also refer to these as “going private transactions” or management buyouts (MBOs). In this paper we focus on buyouts of publicly traded targets in deals sponsored by prominent private equity firms. These deals almost universally involve large amounts of leverage and also frequently involve the target’s pre-deal executive management team. Therefore, we consider all these terms to be practically identical in describing the types of transactions that we are interested in.

cash flow, and stronger managerial incentives (Jensen, 1986). Skeptics suggest that the large profits earned by some private equity partnerships may be partly due to expropriation of target shareholders and stakeholders, or private equity investors, rather than bona fide efficiency improvements (Perry and Williams, 1994; Phalippou, 2009). Other criticisms of LBOs in the 1980s and the 2001–2007 period are that private equity firm profits stem in part from taking advantage of the tax deductibility of interest payments and/or inefficient bond markets in which credit spreads are occasionally excessively low.

Recently, particular criticism has been directed at the so-called club deals, in which two or more private equity firms jointly sponsor an LBO. Clearly, one concern about club deals is that private equity partnerships may be colluding to depress prices by limiting the number of competing bidders in an auction for a takeover target, and thereby may be shortchanging passive, dispersed shareholders of target publicly traded corporations. This concern has strong grounding in the auction literature, in which it is well-recognized that bidder collusion may depress sale prices (e.g., Graham and Marshall, 1989; Marquez and Singh, 2009), and in the regulatory economics literature (e.g., Cramton and Schwartz, 2000; Hendricks and Porter, 1992). These literatures stress that collusion can reduce prices even in the absence of repeat play and even if collusion does not involve all potential bidders for a target. Another, more innocuous, concern is that the limited number of private equity firms interested in a given target may mean that clubbing, even when done in the absence of collusion and for benign reasons, may lead to a reduction in bid competition and hence to lower premiums for target shareholders.

In the absence of an instrumental variable for club formation, it is, of course, not possible to conclusively prove any causal effect of club deals on premiums paid to target shareholders (Bailey, 2007). Even if such an instrument were available, it is also impossible to explicitly distinguish between deliberate collusion to reduce prices and inadvertent reduction in bid competition resulting from the limited number of private equity firms interested in any given target, because intent is unobservable. Nor is it possible to conclusively prove whether club deals result in increased or decreased social welfare.

That said, it is possible to explore the association between club deals and premiums paid to target shareholders, how this association varies in the time-series and cross-section, whether there exist differences in observable characteristics between target firms in club deals and those in other types of deals, and also to consider possible interpretations to advance our understanding of club deals. It is also possible to investigate whether any differences in premiums paid in club deals relative to other types of deals are likely to be due to characteristics suggested by many of the classic theories of deal syndication.

The classic theories of deal syndication suggest a number of benign reasons, unrelated to competition, for why private equity firms may syndicate deals by forming clubs. Capital constraints may induce the formation of clubs for sufficiently large transactions. Even if capital constraints do not bind, diversification motives may

induce funds to syndicate sufficiently large or risky deals. Club deals may also be motivated by a desire to certify deal quality to debt financiers. LBOs are highly levered, and it may be easier to acquire debt financing in sufficient quantity and on favorable terms if multiple private equity firms attach their names and reputations to a deal.

In this paper we provide evidence on these explanations for, and concerns about, club deals in leveraged buyouts by examining the pricing and characteristics of club deals relative to sole-sponsored LBOs and to other merger and acquisition (M&A) transactions. In doing so, we add to the evidence on the differential pricing and characteristics of acquisitions conducted by private equity firms compared to those conducted by other types of acquirers (Bargeron, Schlingemann, Stulz, and Zutter, 2008). Concerns about club deals in the press, and elsewhere, are naturally focused on prominent (or large) private equity firms, because minor private equity firms are less likely to have the market power to meaningfully reduce competition and therefore prices (inadvertently or otherwise) by forming clubs. Accordingly, we conduct our analysis using a comprehensive sample of completed LBOs of U.S. publicly traded targets conducted by prominent private equity firms between January 1984 and September 2007.²

Our main finding is that target shareholders in club deals receive significantly lower premiums than in sole-sponsored LBOs and other merger and acquisition transactions. The differences are economically large: target shareholders receive approximately 10% less of pre-bid firm equity value, or roughly 40% lower premiums, in club deals compared to sole-sponsored LBOs.³ These results are robust to controls for target and deal characteristics, including size, Q, measures of risk, and time and industry fixed effects. The results are also robust to private equity firm fixed effects, indicating that, even though the tendency to club together is concentrated amongst a few prominent private equity firms, our results are not driven by the ability of a few private equity partnerships to pay low prices regardless of whether they act alone or as part of a club. Our results also hold in regression and matching analyses using an LBO-only sample.

This club deal discount is concentrated in club deals announced *prior* to 2006; the financial media began expressing concerns about club deals at the end of 2005 and the U.S. Department of Justice started an informal inquiry into the practice in 2006.⁴ In fact, the club deal

² We define prominent private equity firms as the largest 50 firms by worldwide fundraising over 2002–2007, as well as the private equity arms of major investment banks, and private equity firms that were prominent in the 1980s but are now defunct. We describe the sample in detail in Section 3.

³ Depending on the premium measure, time period, and estimation technique, our estimates of the club deal discount range from 5% to 24% of pre-bid target equity or firm value, with most estimates in the range of 8–12%.

⁴ See Andrew Sorkin, *New York Times*, October 16, 2005, and “Private equity firms face anticompetitive probe,” *The Wall Street Journal*, October 10, 2006. A competing explanation for this structural break is the increased inflow of funds to the private equity industry around the same time, but we do not find supporting evidence for this. Even so, it is, of course, not possible to conclusively attribute any structural break in time to a specific event with time-series evidence alone.

discount virtually disappears in 2006 and 2007. In addition, the intensity of club deals relative to overall LBO activity declines in 2006–2007 compared to 2004–2005.

In the cross-section, the discount associated with club deals is less pronounced for target firms with higher institutional ownership. This result is incremental to the positive relation between institutional ownership and premiums in the overall sample. Notably, there is no similar incremental effect for sole-sponsored LBOs. These results suggest that institutional investors are able to bargain effectively with clubs, and can at least partially counteract the pricing consequences of any reduction in competition (inadvertent or otherwise) associated with clubs. The lack of a relation in sole-sponsored deals may reflect ineffective bargaining or that competitive pressures are strong enough that the bargaining abilities of institutional investors are redundant.

We also examine the robustness of our conclusions to alternative measures of prices paid in private equity sponsored deals. Consistent with the premiums-based evidence, we find that deal values as multiples of sales or earnings before interest, taxes, depreciation, and amortization (EBITDA) (measured relative to multiples paid by acquirers other than prominent private equity firms in time, size, and industry-matched acquisitions) are significantly lower in club deals than in sole-sponsored LBOs prior to 2006. Our results using returns-based measures of premiums are also robust to defining premiums as the final bid price scaled by the pre-announcement target share price.

To further investigate the notion that our findings of lower target premiums in club deals are due to club formation reducing competition for target firms (inadvertently or otherwise), we examine the extent to which the target firm receives competing bids after announcement of the private equity takeover. Consistent with clubs reducing competition for target firms, we find that even controlling for target size, there are significantly fewer post-announcement competing bids in successful club deals than in successful sole-sponsored LBOs. In addition, club targets – but not sole-sponsor targets – with high institutional ownership receive more post-announcement competing bids. These findings suggest that soliciting competing bids *ex post* – thereby forcing the club to raise its price – is one mechanism by which institutions mitigate the discount otherwise associated with club deals.

We next investigate directly whether the benign rationales for club formation outlined above appear to be important motivations for club formation. Of course, we are only able to investigate benign motivations for club formation that are related to observable target characteristics, and cannot rule in or out motivations based on target characteristics that are unobservable to the econometrician.

Regarding capital constraints, the evidence is mixed. We find that club deals are significantly larger on average than sole-sponsored LBOs, which is *prima facie* evidence that capital constraints may be a motivation for club deals. Examining the distribution more closely, however,

we find that the size difference is not statistically significant in the pre-2006 period (in which we find that club targets receive significantly lower premiums). Moreover, only 23% of club deals are larger than the largest sole-sponsored LBO conducted by any of the club members in the four-year window centered on the club deal announcement date.⁵ These facts cast doubt on the notion that capital constraints are a primary motivation for the majority of club deals, especially before 2006, although it is possible that the degree of capital constraints faced by certain private equity firms may be dependent on deal characteristics that we cannot observe.

Regarding diversification motives, we find no evidence that club deal targets are systematically riskier or harder to value as measured by beta, historical stock-return volatility, historical cash flow volatility, number of business segments (a measure of complexity), or analyst forecast errors (a measure of information asymmetry) than targets of sole-sponsored LBOs are.

Moreover, we find that debt financing terms are insignificantly better in club deals than they are in sole-sponsored LBOs. If club deal target firms are riskier, we would expect to observe worse financing terms. In fact, we do not find statistically significant differences in financing terms, measured by loan spreads and maximum debt-to-EBITDA covenants, between club and sole-sponsored LBOs.^{6,7} These results also partially address the concern that club targets receive lower prices because they are riskier in ways that are unobservable to us as econometricians but observable to market participants.

Overall, we interpret these results as providing only limited support for the notion that capital constraints, diversification motives, or financing terms are first-order motivations for club deals.

Given the potential legal and regulatory implications of the possibility of collusion by prominent private equity firms, it is important to emphasize that the patterns we show in the data are not necessarily due to collusion. Indeed, all of our evidence is potentially consistent with clubs forming for (possibly unobservable) benign reasons, with inadvertent pricing consequences.

In addition, because there does not exist a credible instrument in this setting, it is also possible that there are unobservable factors that both drive club formation and can explain the level pricing result, the 2006 break, and the cross-sectional relation between club premiums and target institutional ownership, even in the absence of any reduction in competition. However, it seems difficult to explain our results with simple unobservable aspects of deal risk or quality.

⁵ Our main results are qualitatively unaffected if we drop these club deals (16 in total), which may have been too large to have been sole-sponsored.

⁶ Interestingly, we find that club deals involve significantly more lenders than sole-sponsored LBOs, which may constrain the supply of debt financing for competing bids to the extent that lenders are asked to be exclusive and commit to not lend to other potential bidders for the same target.

⁷ These results are consistent with Axelson, Jenkinson, Strömberg, and Weisbach (2007), who find that debt-to-EBITDA ratios are insignificantly higher in club deals.

Our results on debt terms, described above, suggest that club deals are likely not riskier in unobservable ways that would be reflected in debt terms. Moreover, Guo, Hotchkiss, and Song's (2009) results suggest that it is unlikely that our finding of a pre-2006 club deal discount can be explained by target firms in club deals being of lower quality. Specifically, they find that the ex post performance of leveraged buyouts in the 1990–2006 period is higher in club deals than in sole-sponsored deals, both in terms of return on the price paid, and in terms of accounting performance, suggesting that club deal target firms are of higher quality (i.e., offer higher potential returns) than sole-sponsor targets are.⁸

Another possibility is that coordination costs amongst club members lower a club's willingness to pay. This effect could explain a club deal discount if capital constraints bind and so make a club the only viable acquirer. However, as noted above, the data offer little support for the notion that capital constraints are a primary motivation for the majority of club deals, especially before 2006, and our pricing results are robust to excluding the club deals that are the most likely to be driven by capital constraints.

Our analysis does not speak to the social welfare or efficiency implications of club deals. Club deals may be economically efficient and socially beneficial if they redeploy assets to more productive uses that would remain undiscovered without the pooling of multiple expert opinions, even if there is room for target shareholders to share more of the surplus. Several studies of the economic impact of private equity show substantial long-term benefits to society (Gurung and Lerner, 2008). Whether club deals are different from sole-sponsored LBOs along this dimension is an important question for future research.

Finally, we note that our results on the pricing of club deals differ from those of Boone and Mulherin (2009) and Barger, Schlingemann, Stulz, and Zutter (2008). Boone and Mulherin (2009) examine the pricing of club deals in the 2003–2007 period, but, unlike us, find no evidence of a club deal discount in long-horizon acquisition premiums. Similarly, Barger, Schlingemann, Stulz, and Zutter (2008, footnote 4) report but do not tabulate that they find an insignificant club deal discount using long bid-period windows in a sample of 43 club deals conducted over 1980–2005. Neither of these studies distinguishes between prominent and other private equity firms, nor accounts for the important structural break in 2006 that we find (the Barger et al. sample ends in 2005). Our results combine with theirs to suggest that the pre-2006 club deal discount we show is likely specific to large, prominent private equity firms. Given the nature of the concerns about club deals, we believe our focus on prominent private equity firms is a distinguishing strength of our work, both because these private equity firms are the most likely to have the market power to constrain competition (inadvertently or otherwise) with

meaningful pricing consequences, and because they account for the vast majority of dollar deal volume and fundraising in the private equity industry.

Our paper proceeds as follows. Section 2 presents relevant related literature. Section 3 describes our sample selection procedure and provides summary statistics on the resulting sample. Section 4 analyzes the gains to target shareholders using stock-return measures. Section 5 considers multiples-based pricing measures. Section 6 provides further evidence on post-announcement competition and on benign motivations for club deals. Section 7 concludes.

2. Related literature

Our work adds to the literature on the pricing and characteristics of LBOs and going private transactions, and the possible sources of returns to private equity partnerships. DeAngelo, DeAngelo, and Rice (1984) examine a sample of 81 going private transactions announced between 1973 and 1980, and conclude that the average going private transaction does not result in expropriation of wealth from dispersed target shareholders. Kaplan (1989a) investigates pre- to post-buyout changes in operating performance and market value for management buyouts conducted in the 1980s that subsequently went public and finds evidence consistent with substantial efficiency improvements. Kaplan (1989b) finds that the tax deductibility of interest payments is an important source of the wealth gain in buyouts. Kaplan and Stein (1993) examine the LBO wave of the 1980s, and find evidence consistent with “overheating” of the LBO market towards the end of the wave, when credit spreads may have been inefficiently low. Perry and Williams (1994) find that target managers appear to manipulate earnings downward in the year preceding a management buyout (using discretionary accruals), consistent with complicity in lowering the acquisition price, although DeAngelo (1986) finds contradictory evidence. Despite these possible sources of returns, Phalippou (2009) argues that investors in private equity funds earn low risk-adjusted returns.

Club deals in leveraged buyouts have interesting parallels to the syndication of venture capital (VC, or early-stage private equity) investments. While a number of papers examine syndication motives in VC, to our knowledge we are the first to provide direct evidence on the empirical relevance of possible syndication motives in the LBO market. In the VC setting, Lerner (1994) and Du (2008) provide evidence that syndication more commonly occurs between VC firms of similar prominence, at least in first-round investments, which mirrors our findings. Brander, Amit, and Antweiler (2002), among others, find support for diversification motives in VC syndication, whereas we find little support for those motives in our setting.

With respect to deal pricing, Hochberg, Ljungqvist, and Lu (2009) find that syndication networks among venture capital firms are associated with low valuations for entrepreneurs seeking financing, i.e., they find that VC

⁸ Similarly, Brander, Amit, and Antweiler (2002) find that syndicated venture capital (VC) investments outperform non-syndicated ones.

syndication is associated with lower deal prices, but cannot distinguish between benign and other explanations. Their findings and conclusions are consistent with ours.

Finally, our work is also part of a nascent literature in “forensic finance.” Ritter (2008) provides an excellent discussion of this literature, emphasizing the important role of statistical analysis in assessing whether patterns in financial market data are likely to have innocuous explanations.

3. Sample

Our sample of mergers and acquisitions comes from Thomson Financial’s Securities Data Company (SDC) Platinum database. We extract all transactions announced between January 1984 and September 2007 that are completed by the end of November 2007, in which the deal value is greater than \$100 million, the deal is characterized by SDC as a merger, acquisition, or LBO, and the acquirer is seeking to own more than 50% of the target’s shares outstanding.⁹ To ensure that we obtain a sample of publicly traded target firms, we match this preliminary sample of 14,335 transactions to the Center for Research in Security Prices (CRSP) database, and require that the target firm have a stock price in the CRSP database at least once in the seven days prior to the announcement date reported by SDC. Furthermore, to ensure that all transactions in the data set are completed acquisitions, we require a delisting event within one year (most are within one month) of the deal effective date in SDC. This procedure leaves 4,031 completed acquisitions of publicly traded target firms.

Because the concerns about club deals primarily focus on large buyouts by prominent private equity firms, we determine which of these transactions are LBOs led by prominent private equity firms. It is not obvious how to do this or how to determine whether any given LBO is a club deal. While SDC contains an LBO flag and a “going private” flag, it is unclear what criteria are used to set these flags. We observe that the flags capture some transactions that are not sponsored by private equity firms, such as corporation-led buyouts and management buyouts that do not involve a private equity firm, and also miss some transactions that are sponsored by private equity firms. In fact, the LBO flag provided by SDC misses about one in five of the deals we identify using the procedure outlined below.

In light of these difficulties, we code indicator variables for whether a private equity firm is involved in the deal and for whether multiple private equity firms are involved. This coding is done by hand for a sample of acquisitions that we identify using a text search for the names of prominent private equity firms in the transaction synopses for the sample of 4,031 deals from SDC. Unfortunately

there is no data source that we know of that provides a comprehensive list of prominent private equity firms that have ever been active in the U.S. public-to-private market. We create our own list in the following way.

We begin with the names of the 50 largest private equity firms in the world as reported in the May 2007 issue of *Private Equity International* (PEI) magazine. PEI’s ranking is based on capital raised over the five-year period between 2002 and 2007, and this list of prominent private equity firms contains all the well-known financial sponsors of LBOs.¹⁰ To this list of prominent LBO sponsors we add the names of the in-house private equity units of the investment banks Merrill Lynch, Morgan Stanley, and JP Morgan (including Chase Capital Partners). Despite their importance, these sponsors may not be on the PEI list because they may use internal capital rather than relying on external fundraising. We also add Forstmann Little and HM Capital Partners (formerly Hicks, Muse, Tate, and Furst) because they are historically prominent LBO sponsors that have been less active in recent fundraising.

If a deal synopsis from SDC contains the name of one of the prominent private equity firms described above, and the role of that firm is an acquirer, then we code that deal as a private equity deal. Further, if that deal synopsis also contains the name of another private equity firm (even a minor one) as a joint acquirer, then the deal is coded as a “club” deal. While there is some discretion involved in hand-coding transactions based on text synopses from SDC, we conduct the search in a conservative way that is intended to ensure that all deals coded as sponsored by private equity firms are, in fact, such transactions. The resulting sample consists of 201 private equity deals of which 70 are club deals. The Appendix contains a list of the private equity deals in our sample, with year, target name, and acquirer name (truncated at two acquirer names for club deals) tabulated.¹¹ Panel A contains the list of club deals in our sample, and sole-sponsored private equity deals are in Panel B. From this point onwards in this paper, when we refer to private equity deals (sole-sponsored or club) we are referring to deals involving these prominent private equity partnerships.

Table 1 displays the distribution of our sample by year and acquirer type. We consider four mutually exclusive and exhaustive acquirer types: private equity clubs, sole private equity firms, private firms that are not prominent private equity firms (therefore, this category includes not only private firms but also non-prominent private equity firms), and public firms. The merger waves of the late

¹⁰ Highlighting the importance of focusing on prominent private equity sponsors, PEI reports that the members of its top-50 list were responsible for 75% of global LBO deal volume during 2002–2007.

¹¹ We thank one of our anonymous referees and several colleagues for pointing out several sample inaccuracies that resulted from the sole use of SDC data for classification. We decided to classify both the 2007 CDW and 1999 Concentra Managed Care LBOs as sole-sponsored deals because the mode of club formation in both deals was unusual (both were sole-sponsored deals at announcement). However, all our results are robust to reclassifying these deals as Club LBOs. Details of the sample classifications that would not be replicable using SDC data (but which are reflected in the Appendix) are available from the authors by request.

⁹ Our main results are qualitatively unaffected if we instead impose a real deal value screen of \$100 million in 2007 dollars. We use a nominal cutoff of \$100 million to be consistent with prior literature, even though this results in more stringent criteria for inclusion in the sample in 1984 than in 2000 or 2007 (because of the dramatic rise in the level of stock market prices around 2000 and 2007).

Table 1

Sample distribution. This table contains the time-series distribution of a sample of acquisitions of publicly traded targets from SDC. The transactions are classified into years based on announcement dates. We identify private equity (PE) deals by performing a text search for the names of prominent private equity firms in the deal synopses provided by SDC and manually confirming the results. Prominent private equity firms are defined as the 50 largest private equity firms in the world as reported in the May 2007 issue of *Private Equity International* (PEI) magazine (with five additions described in Section 3). If the synopsis from SDC contains the name of *only one* prominent private equity firm (and no other firms are identified as the acquirer), then the deal is classified as a Sole PE deal. If the synopsis from SDC contains the name of *at least one* prominent private equity firm (and at least one other private equity firm, prominent or not, is identified as the acquirer), then the deal is classified as a Club deal. In a few cases the SDC synopses are incorrect or misleading, and we use merger proxy statements to classify these deals (see Section 3 for details). The remaining transactions in the sample are classified by whether the acquirer is privately held (Private bidder deals) or publicly traded (Public bidder deals), with listing status determined by the bidder's inclusion in the CRSP database. Total deal value is the annual sum of deal value excluding assumed liabilities reported by SDC (in \$m).

Year	All deals		Deals by prominent private equity firms				Other deals			
			Club deals		Sole PE deals		Private bidder		Public bidder	
	No. of deals	Total deal value	No. of deals	Total deal value	No. of deals	Total deal value	No. of deals	Total deal value	No. of deals	Total deal value
1984	33	17,359.1	0	0.0	2	882.2	13	5,809.7	18	10,667.2
1985	111	96,555.4	1	637.0	5	11,696.8	19	8,190.4	86	76,031.2
1986	131	79,560.9	2	1,381.0	8	12,387.4	41	25,642.2	80	40,150.3
1987	127	73,839.8	1	103.9	7	11,973.0	49	25,310.0	70	36,452.9
1988	135	127,241.3	1	538.9	10	34,013.0	74	49,630.0	50	43,059.4
1989	103	89,764.5	1	923.0	2	2,204.1	48	24,679.9	52	61,957.5
1990	49	41,629.3	0	0.0	3	2,500.3	17	7,644.1	29	31,484.9
1991	47	29,686.7	0	0.0	0	0.0	3	2,508.9	44	27,177.8
1992	54	24,431.9	0	0.0	0	0.0	3	998.8	51	23,433.1
1993	74	67,274.8	1	191.3	2	497.7	4	1,259.7	67	65,326.1
1994	130	88,392.1	0	0.0	1	2,243.4	25	17,825.6	104	68,323.1
1995	181	180,040.4	2	1,642.3	2	1,357.2	24	28,171.6	153	148,869.3
1996	220	247,262.4	1	112.8	5	2,334.2	31	33,570.2	183	211,245.2
1997	353	332,194.3	3	2,187.4	11	5,277.2	51	23,106.9	288	301,622.8
1998	351	960,167.2	5	2,691.2	3	1,352.8	64	43,783.0	279	912,340.2
1999	402	829,847.4	5	2,437.0	7	3,478.2	95	73,128.2	295	750,804.0
2000	330	857,332.8	2	613.7	3	1,783.2	83	85,100.9	242	769,835.0
2001	187	276,961.3	0	0.0	1	995.0	43	52,786.5	143	223,179.8
2002	112	141,999.3	3	2,302.5	2	741.7	28	8,521.7	79	130,433.4
2003	141	201,788.7	0	0.0	2	490.7	21	10,501.6	118	190,796.4
2004	163	356,586.9	5	12,850.6	6	8,000.6	22	51,857.5	130	283,878.2
2005	199	493,836.1	9	28,517.7	10	10,331.3	55	62,955.3	125	392,031.8
2006	240	633,778.8	19	124,309.1	19	48,388.2	61	57,105.7	141	403,975.8
2007	158	385,159.0	9	83,868.2	20	103,054.5	48	60,546.1	81	137,690.2
Total	4,031	6,632,690.4	70	265,307.6	131	265,982.7	922	760,634.5	2,908	5,340,765.6

1980s and late 1990s are evident in the table. Table 1 also shows that club deals are a relatively new phenomenon: 37 of the 70 take place in 2005–2007, and 63 of the 70 take place in 1995–2007.

Interestingly, the intensity of club deals relative to overall LBO activity declines slightly in 2006–2007 (especially 2007) compared to 2004–2005 (following the onset of media and government scrutiny of the practice). Specifically, there are 30 LBOs in our sample in 2004–2005, of which 14 (47%) are club deals. For 2006–2007, there are 67 LBOs in our sample, of which 28 (42%) involve clubs of private equity firms. While one could argue that this trend is mechanical, because the number of prominent private equity firms is fixed while the number of non-prominent firms likely increases over time, the trend prior to 2004–2005 suggests otherwise. In 2002–2003, there are seven LBOs in our sample, of which three (43%) are club deals. Furthermore, in 2000–2001, there are six LBOs in our sample, of which two (33%) are club deals. This suggests that the drop in club deal intensity in 2006

and especially 2007 is actually a reversal of an increasing trend towards clubbing in the years before that.

Panel A of Table 2 gives a frequency distribution of deals by private equity firm. Kohlberg Kravis Roberts (KKR) leads the table with 26 deals (11 of which are club deals), followed by Blackstone with 24 deals (12 of which are club deals), and TPG with 21 deals (18 of which are club deals, the vast majority of TPG's LBO activity). Participation in club deals roughly mimics overall participation in LBO activity, although there are a few exceptions such as TPG.

Panel B of Table 2 displays a participation matrix, documenting the frequency with which private equity firms partner with each other in club deals. TPG (the most common club participant in Panel A) partners four times each with the other three most prominent private equity groups in our sample (Blackstone, KKR, and Goldman Sachs). Panel B shows that the most prominent private equity firms (who are also the subject of the litigation referred to in the introduction) are more likely to partner with each other than with less prominent private equity

Table 2

LBOs completed by private equity firms: counts and cross-participation. Panel A contains the number of deals sponsored by the most active private equity firms in our sample. Sole PE and Club deals are defined in the legend to Table 1. Panel B presents the number of times prominent private equity firms have undertaken club deals with one another (as opposed to other, non-prominent private equity firms). Our sample contains club LBOs by 29 prominent private equity firms of which Panel B lists 24. The remaining five prominent private equity firms, Berkshire Partners, Cerberus Capital Management, Clayton, Dubilier & Rice, Fortress Group, and Sun Capital Partners, are not listed because although Cerberus has done two club deals and the others have done one club deal each, none of them are with other prominent private equity firms. In Panel B, columns that do not contain additional information are omitted to save space.

Panel A: Number of deals												
Rank	PEI 50	Name of private equity firm						All deals	Sole PE deals		Club deals	
1	2	Kohlberg Kravis Roberts						26	15		11	
2	4	The Blackstone Group						24	12		12	
3	5	TPG						21	3		18	
4	3	Goldman Sachs Principal Investment Area						18	5		13	
5	42	Welsh, Carson, Anderson & Stowe						14	9		5	
6	12	Apollo Management						13	9		4	
7	...	Morgan Stanley						12	10		2	
8	...	Merrill Lynch						10	6		4	
9	30	Thomas H. Lee Partners						10	4		6	
10	31	Leonard Green & Partners						9	6		3	
11	1	The Carlyle Group						9	5		4	
12	...	Forstmann Little						8	8		0	
13	8	Bain Capital						8	2		6	
14	...	HM Capital Partners (formerly Hicks, Muse, Tate, and Furst)						7	6		1	
15	32	Madison Dearborn Partners						7	3		4	
16	14	Warburg Pincus						6	4		2	
17	9	Providence Equity Partners						6	0		6	
18	47	Clayton, Dubilier & Rice						5	4		1	
19	25	Lehman Brothers Private Equity						5	4		1	
20	19	Silver Lake Partners						5	2		3	
21	16	Hellman & Friedman						5	1		4	
22	...	JP Morgan (including Chase Capital Partners)						5	1		4	
23	27	Fortress Investment Group						4	3		1	
24	34	Cerberus Capital Management						3	1		2	
25	44	GTCR Golder Rauner						3	1		2	
26	40	Berkshire Partners						2	1		1	
27	33	Onex						2	1		1	
28	28	Sun Capital Partners						2	1		1	
29	39	TA Associates						2	1		1	
30	29	BC Partners						1	1		0	
31	21	EQT Partners						1	1		0	
32	41	Pacific Equity Partners						1	1		0	
33	6	Permira						1	0		1	
Panel B: Cross-participation matrix												
	TPG	The Blackstone Group	Kohlberg Kravis Roberts	Goldman Sachs	Providence Equity Partners	Bain Capital	Thomas H. Lee Partners	The Carlyle Group	JP Morgan	Morgan Stanley	Merrill Lynch	GTCR Golder Rauner
The Blackstone Group	4	...	...	...	...	...	...	...	...	...	...	...
Kohlberg Kravis Roberts	4	3	...	...	...	...	...	...	...	...	...	...
Goldman Sachs	4	4	4	...	...	...	...	...	...	...	...	...
Providence Equity Partners	3	1	2	2	...	...	...	...	...	...	...	...
Bain Capital	1	2	3	1	1	...	...	...	...	...	...	...
Thomas H. Lee Partners	1	0	0	1	1	0	...	...	...	...	...	...
The Carlyle Group	1	1	1	1	2	0	0	...	...	...	...	...
JP Morgan	0	0	0	1	0	0	1	0	...	...	...	...
Morgan Stanley	1	0	1	1	0	0	0	0	0	...	...	...
Merrill Lynch	0	0	1	0	0	1	1	0	0	0	...	...
GTCR Golder Rauner	0	0	0	0	0	0	0	0	0	0	0	...
Silver Lake Partners	3	1	1	1	1	1	0	0	0	0	0	0

Table 2 (continued)

Panel B: Cross-participation matrix												
	TPG	The Blackstone Group	Kohlberg Kravis Roberts	Goldman Sachs	Providence Equity Partners	Bain Capital	Thomas H. Lee Partners	The Carlyle Group	JP Morgan	Morgan Stanley	Merrill Lynch	GTCR Golder Rauner
Lehman Brothers Private Equity	1	0	1	1	0	0	0	0	0	1	0	0
Madison Dearborn Partners	1	0	0	0	1	0	1	0	0	0	1	0
Hellman & Friedman	2	1	1	0	0	0	0	0	0	0	0	0
Warburg Pincus	1	0	0	1	0	0	1	0	1	0	0	0
Welsh, Carson, Anderson & Stowe	0	1	1	0	0	0	0	0	0	0	0	1
Permira	1	1	0	0	0	0	0	1	0	0	0	0
Leonard Green & Partners	2	0	0	0	0	0	0	0	0	0	0	0
Apollo Management	0	0	0	0	0	0	0	0	1	0	0	0
HM Capital Partners	0	0	1	0	0	0	0	0	0	0	0	0
TA Associates	0	0	0	0	0	0	0	0	0	0	0	1
Onex	0	0	0	0	0	0	0	0	0	1	0	0

firms. This finding is consistent with the *prima facie* concern about collusion lowering deal prices: the most prominent private equity firms, with the market power to effectively collude and, in all likelihood, the greatest ability to compete against one another absent collusion, are the most likely to partner in LBO club deals.

4. Returns to target shareholders

In this section, we use several measures of the gains to target shareholders in our sample to provide evidence on the pricing of club deals relative to sole-sponsored LBOs and relative to non-private equity transactions.

4.1. Variable construction

We collect daily return data for each of our targets from CRSP. We use these data to construct the raw returns, buy-and-hold abnormal returns (BHARs), and cumulative abnormal returns (CARs) over the following intervals for each target (all dates are *trading* days relative to the deal announcement date (0)): day -42 to day -1 (labeled *runup*); day 0 to day $+126$ or the delisting date, whichever occurs first (labeled *markup*); and day -42 to day $+126$ or the delisting date, whichever occurs first (labeled *premium*). Our choice of intervals follows Schwert (1996). We consider long-run CARs for consistency with some of the existing literature, even though there is a greater potential for pronounced differences between compounded and cumulated returns over longer horizons, and even though BHARs are a better measure of the wealth effects to target shareholders.

All BHARs are calculated by subtracting the compound return to the CRSP value-weighted market index (including

dividend distributions) from the compound return to the target shares over the given period.¹² We calculate all CARs by summing market-model residuals ($\tilde{\varepsilon}_{it}$), and estimate market-model parameters ($\tilde{\alpha}_i$ and $\tilde{\beta}_i$) for each target using daily returns from trading day -379 to trading day -127 relative to the deal announcement date:

$$\tilde{\varepsilon}_{it} = r_{it} - (\tilde{\alpha}_i + \tilde{\beta}_i r_{mt}),$$

where r_{it} is target i 's return on day t and r_{mt} is the return to the CRSP value-weighted market index (including dividend distributions) on day t . In addition to the above-mentioned variables, we also calculate the cumulative abnormal return to the target's shares from trading day -1 to trading day $+1$ (labeled *CAR3*). These return/premium measures are all used in a variety of papers in the M&A literature (see, for example, Schwert, 1996; Barger, Schlingemann, Stulz, and Zutter, 2008).

4.2. Target return comparisons by bidder type

Table 3 displays means and medians of these return measures classified by acquirer type. Our results are easily summarized. For all return measures except *runup*, shareholders of club deal targets earn significantly lower

¹² Our results are robust to defining the buy-and-hold abnormal return as the target's compound return over the relevant period minus the compound return to the matching Fama-French (5×5) size and book-to-market portfolio (style-adjusted buy-and-hold abnormal return) as well as defining bid premiums using SDC deal prices relative to target share prices 43 days prior to announcement (as in Officer, 2003, Table 3). Results using these alternative measures of abnormal returns and premiums are omitted for brevity.

Table 3

Target percentage returns, by bidder type. This table contains averages and medians (in brackets) for various measures of LBO target returns (in percent) in the sample of deals from SDC described in the legend to Table 1. Runup is the compound return measured over the (–42, –1) period (defined as trading days relative to the announcement date (0)). CR3 is the compound return measured over the (–1, +1) period, Markup is the compound return measured over the (0, +126) period (unless the target is delisted prior to trading day +126), and Premium is the compound return measured over the (–42, +126) period. All buy-and-hold abnormal returns (BHAR) are the compound return to the target shares minus the compound return to the CRSP value-weighted market index (including dividend distributions) over the same period. Cumulative abnormal returns (CAR) are the sum of the target's daily raw market-model residuals over the corresponding periods (with market-model parameters estimated over the (–379, –127) period as long as there are 100 returns on CRSP in that period). ***, **, and * indicate that the difference in target returns is significantly different from zero at the 1%, 5%, and 10% levels, respectively.

	Differences									
	Club	Sole PE	Private	Public	Club–Sole PE	Club–Private	Club–Public	Sole PE–Private	Sole PE–Public	Private–Public
Raw returns										
Runup	8.90 [7.88]	12.60 [10.48]	15.23 [10.23]	14.25 [10.48]	–3.70 [–2.60]	–6.33*** [–2.35]	–5.35** [–2.61]	–2.64 [0.25]	–1.65 [–0.01]	0.99 [–0.25]
CR3	11.76 [9.70]	19.12 [15.55]	22.62 [18.13]	20.19 [16.50]	–7.37*** [–5.85]**	–10.86*** [–8.43]***	–8.44*** [–6.80]***	–3.50* [–2.59]	–1.07 [–0.95]	2.43*** [1.64]
Markup	14.21 [13.63]	21.91 [20.65]	26.48 [22.18]	26.61 [22.58]	–7.70** [–7.02]**	–12.27*** [–8.55]***	–12.40*** [–8.96]***	–4.57** [–1.53]	–4.70** [–1.93]	–0.13 [–0.40]
Premium	24.04 [22.97]	36.11 [32.34]	44.10 [36.72]	43.81 [37.44]	–12.07*** [–9.37]**	–20.06*** [–13.75]***	–19.76*** [–14.47]***	–7.99*** [–4.38]	–7.70*** [–5.10]*	0.29 [–0.72]
Number observations	70	131	922	2,908						
Buy-and-hold abnormal returns										
Runup BHAR	6.40 [4.35]	9.24 [6.25]	12.76 [7.63]	11.61 [7.83]	–2.84 [–1.91]	–6.35** [–3.28]	–5.21*** [–3.48]*	–3.51* [–1.37]	–2.37 [–1.58]	1.14 [–0.21]
Markup BHAR	8.08 [7.44]	16.09 [14.41]	22.91 [18.04]	21.97 [18.17]	–8.01** [–6.97]**	–14.83*** [–10.60]***	–13.89*** [–10.74]***	–6.82*** [–3.63]*	–5.88*** [–3.76]*	0.94 [–0.13]
Premium BHAR	15.32 [12.92]	26.82 [23.13]	37.99 [31.02]	36.46 [30.05]	–11.49*** [–10.21]**	–22.67*** [–18.10]***	–21.14*** [–17.13]***	–11.17*** [–7.89]**	–9.64*** [–6.92]**	1.53 [0.97]
Number observations	70	131	922	2,908						
Cumulative abnormal returns										
Runup CAR	7.18 [6.31]	8.45 [6.14]	12.22 [8.02]	9.63 [7.41]	–1.27 [0.17]	–5.04*** [–1.72]	–2.45 [–1.11]	–3.78** [–1.89]	–1.18 [–1.27]	2.59*** [0.61]
CAR3	11.45 [10.10]	18.26 [15.76]	22.28 [17.75]	19.54 [16.23]	–6.81*** [–5.66]**	–10.83*** [–7.66]**	–8.09*** [–6.13]**	–4.02** [–2.0]*	–1.28 [–0.47]	2.74*** [1.53]**
Markup CAR	9.64 [7.02]	14.76 [14.94]	21.40 [18.14]	18.01 [15.55]	–5.12* [–7.92]**	–11.76*** [–11.12]**	–8.38*** [–8.53]**	–6.64*** [–3.20]*	–3.25 [–0.61]	3.39*** [2.59]**
Premium CAR	16.81 [19.18]	23.21 [23.89]	33.62 [28.82]	27.64 [24.89]	–6.39 [–4.71]	–16.81*** [–9.64]***	–10.83*** [–5.71]***	–10.42*** [–4.93]**	–4.43 [–1.01]	5.98*** [3.92]***
Number observations	70	128	902	2,786						

returns than shareholders of targets that are acquired by any other category of acquirer. This is true in both means and medians. The only exception is that for one measure, cumulative abnormal returns (CARs) measured over the premium period, the point estimate of the difference between sole-sponsored and club deals (–6.4%) is not statistically significant at conventional levels, with a *p*-value of 0.13. The economic magnitudes of the differences in target returns between club and sole-sponsor deals are generally large: for example, the differences in the buy-and-hold abnormal returns over the whole bid-period (buy-and-hold abnormal premium) range from 11 to 23 percentage points, on average. Thus, target shareholders gain considerably less from acquisitions by clubs of private equity firms than from acquisitions by other types of acquirers. The insignificant results for the runup measures suggest that pre-announcement information

leakage or deal anticipation is not different in club deals than in other types of deals.¹³

Of course, our main interest is a comparison between club deals and sole-sponsored private equity deals. Even if private equity deals generally are priced lower than deals by publicly traded acquirers, skepticism of club deals may be misplaced if pricing does not differ between the two types of private equity deals.

Table 3 shows that, on average, shareholders of club deal targets earn a buy-and-hold abnormal return over the premium period (–42 to +126 trading days relative to the deal announcement) that is a statistically and

¹³ The sample sizes differ slightly between cumulative and buy-and-hold abnormal returns in Table 3 because the former relies on the estimation of market-model parameters (described in Section 4.1) while the latter does not.

economically significant 11.5 percentage points, or 43% less than shareholders of targets involved in sole-sponsored deals. Another, more striking, way to view the differences between club deals and sole-sponsored private equity deals is that target shareholders gain 75% greater abnormal premiums, on average, if the acquisition is by a sole private equity firm rather than a club (26.8% vs. 15.3%). Table 3 also shows that the result that target shareholders gain less in club deals than in sole-sponsored private equity deals is robust to different return measurement periods and return definitions.

4.3. Differences in target firm characteristics

It is possible that the reason clubs of private equity firms pay less to target shareholders than do other types of acquirers is that they seek to acquire different types of firms. We investigate this possibility by focusing on observable target characteristics that the literature has found to be important in explaining gains to target shareholders, while acknowledging that we are unable to observe differences in dimensions that are unobservable to the econometrician. Table 4 compares target firm

Table 4

Target characteristics, by bidder type. This table contains averages and medians (in brackets) for various target characteristics. Numbers beneath medians are the number of observations. Size is the target's market capitalization 43 days prior to bid announcement (immediately before the beginning of the runup period), measured in billions of dollars. Accounting data for the target firm are from Compustat for the fiscal year ending immediately prior to deal announcement. Industry-adjusted Q is the target's Q ratio (defined in Kaplan and Zingales, 1997) minus the median Q for all firms in the same two-digit SIC code industry in the same year. Industry-adjusted EBITDA/Assets is the target's EBITDA/Assets minus the average EBITDA/Assets for all firms in the same two-digit SIC code industry in the same year. Debt/(Debt+Equity) is defined as the book value of the target's total debt (Compustat annual data item #9 plus Compustat annual data item #34) divided by the sum of the book value of the target's total debt and the target's market capitalization 43 days prior to bid announcement. Institutional ownership is measured in the quarter prior to deal announcement, and is the fraction of the target's shares owned by institutions required to file a 13F statement and included in Thomson Financial's 13F Holdings database. Prior 12-month return is the compound return to the target's stock over the one year immediately preceding the beginning of the runup period (i.e., ending on trading day –43 relative to the announcement date), in percent. Prior 12-month BHAR (buy-and-hold abnormal return) is Prior 12-month return minus the compound return to the CRSP value-weighted market index (including dividend distributions) over the same period, in percent. Prior 12-month return volatility is the standard deviation of the target's daily percent returns for the one year immediately preceding the beginning of the runup period. Beta is the target's market-model beta estimated over the (–379, –127) trading day relative to the announcement date period (using daily data) as long as there are 100 returns on CRSP in that period (otherwise Beta is set to missing). ***, **, and * indicate that the difference in averages or medians is significantly different from zero at the 1%, 5%, and 10% levels, respectively.

	Differences									
	Club	Sole PE	Private	Public	Club – Sole PE	Club – Private	Club – Public	Sole PE – Private	Sole PE – Public	Private – Public
Size (\$ billions)	2.81 [0.93] 70	1.36 [0.53] 131	0.54 [0.20] 922	1.28 [0.26] 2,903	1.45** [0.40]*	2.27*** [0.73]***	1.54*** [0.67]***	0.82*** [0.33]***	0.08 [0.27]***	–0.74*** [–0.06]***
Industry-adjusted Q	0.06 [–0.01] 62	0.01 [–0.12] 117	0.00 [–0.09] 821	0.44 [0.01] 2,620	0.05 [0.11]	0.06 [0.08]	–0.38*** [–0.02]	0.01 [–0.03]	–0.43*** [–0.14]**	–0.44*** [–0.10]***
Industry-adjusted EBITDA/assets	0.04 [0.02] 64	0.05 [0.03] 114	0.03 [0.02] 813	0.02 [0.01] 2,639	–0.01 [–0.01]	0.02 [–0.00]	0.03** [0.01]**	0.03*** [0.01]	0.04*** [0.02]**	0.01** [0.01]***
Debt/(debt+equity)	0.28 [0.28] 68	0.30 [0.29] 124	0.30 [0.27] 847	0.26 [0.21] 2,712	–0.02 [–0.02]	–0.02 [0.01]	0.02 [0.07]	0.00 [0.02]	0.04* [0.08]**	0.04*** [0.06]***
Institutional ownership	0.65 [0.69] 62	0.57 [0.56] 110	0.43 [0.42] 814	0.43 [0.41] 2,669	0.08** [0.14]**	0.22*** [0.28]***	0.22*** [0.28]***	0.14*** [0.14]***	0.14*** [0.14]***	0.00 [0.00]
Prior 12-month return	12.73 [4.86] 70	14.10 [11.80] 131	13.41 [7.10] 922	21.94 [11.68] 2,903	–1.37 [–6.94]	–0.69 [–2.24]	–9.21 [–6.82]	0.69 [4.70]	–7.84** [0.12]	–8.53*** [–4.58]**
Prior 12-month BHAR	–2.84 [–8.94] 70	–0.49 [–2.85] 131	–0.57 [–6.87] 922	6.15 [–4.71] 2,903	–2.35 [–6.09]	–2.26 [–2.07]	–8.99* [–4.23]	0.09 [4.02]	–6.64* [1.86]	–6.73*** [–2.17]
Prior 12-month return volatility	2.41 [2.03] 70	2.46 [2.26] 129	3.04 [2.62] 917	3.11 [2.72] 2,855	–0.05 [–0.23]*	–0.62*** [–0.59]***	–0.69*** [–0.70]***	–0.58*** [–0.36]***	–0.65*** [–0.47]***	–0.07 [–0.11]
Beta	0.91 [0.89] 70	0.91 [0.94] 128	0.75 [0.69] 902	0.82 [0.72] 2,786	0.00 [–0.05]	0.16*** [0.19]***	0.09* [0.17]***	0.16*** [0.25]***	0.09** [0.23]***	–0.07*** [–0.02]

characteristics across acquirer types. Given our interest in club deals, we focus mostly on differences between club and sole-sponsored private equity deals.

We begin with target size, measured as market value of equity (from CRSP) 43 trading days prior to the announcement (i.e., the day before the beginning of the runup period). Consistent with the notion that club deals are a way for private equity firms to share the risk inherent in large transactions, club deal targets are significantly larger than targets of other types of acquirers, including targets of sole-sponsored private equity deals. This is true in both means and medians. This fact is potentially important for the interpretation of the results in Table 3 that club deals produce lower premiums for target shareholders in light of existing evidence that target shareholder gains are negatively related to target size (e.g., Comment and Schwert, 1995; Officer, 2003).

We also consider the Tobin's *Q* ratio of the target (defined as in Kaplan and Zingales (1997) as the market value of assets divided by the book value of assets, where the market value of assets equals the book value of assets plus the market value of common equity less the sum of the book value of common equity and balance sheet deferred taxes). Lang, Stulz, and Walking (1989) argue that acquirers have greater scope to create value if low target *Q* (or market-to-book) is caused by rectifiable agency problems at the target firm. Our results confirm the well-known finding (e.g., Barger, Schlingemann, Stulz, and Zutter, 2008) that private equity firms in general acquire significantly “cheaper” targets (lower industry-adjusted *Q* ratios; industries are defined using two-digit Standard Industrial Classification (SIC) codes) than publicly traded acquirers do. However, there is no significant difference in average target industry-adjusted *Q* between club deals and sole-sponsored private equity deals. To the extent that the scope to create value in private equity transactions stems from existing operational inefficiencies that would be reflected in low *Q*, these results suggest that lack of scope to create value is unlikely to explain the lower premiums associated with club deals.

We also examine the target's performance in terms of operating cash flow as measured by the ratio of EBITDA to total assets (from Compustat) for the fiscal year prior to the deal announcement—EBITDA is the most often used measure of cash flow in the private equity literature. There is no significant difference in pre-deal industry-adjusted operating performance for targets of club versus sole private equity firm deals.

We next consider a measure of pre-transaction target leverage, the ratio of the book value of total debt (short- and long-term, from Compustat) to the sum of the book value of debt and the market value of equity. There are no significant differences in pre-deal target leverage between club deal and non-club deal transactions.

Institutional ownership of the target firm may reduce the ability of private equity firms to acquire such firms cheaply. In Table 4 we show the average and median fraction of target shares that are reported as owned by institutional investors in the calendar quarter prior to deal

announcement. Our measure of institutional ownership is from Thomson Financial's 13F Holdings database, which contains the stock-level holdings of institutional money managers (but not hedge funds) collected from mandatory quarterly filings with the Securities and Exchange Commission (SEC). We compute the fraction of a firm's outstanding shares owned by all institutions at the end of the quarter immediately prior to the acquisition announcement date. Targets in club private equity transactions have significantly higher institutional ownership than targets of other acquirers, including sole private equity firms, both on average and at the median. Since these targets are larger firms (on average), this is not surprising.

Lastly, we consider stock-return-based measures of pre-deal performance and risk. We measure pre-deal stock-return performance using buy-and-hold stock returns for the 12 months prior to the beginning of the runup period (trading day –294 to day –43 relative to the announcement date), both raw and market-adjusted. Targets of club acquisitions have generally insignificantly different pre-deal stock-return performance than targets of other acquirers. There is some evidence that publicly traded bidders acquire targets with better pre-deal performance, suggesting that publicly traded acquirers are more likely to target well-performing and highly valued firms than private equity groups (or private firms) are.

We measure pre-deal target risk in two ways: using the standard deviation of the target's daily returns over the 12 months prior to the beginning of the runup period and using the beta estimated in the market model described in Section 4.1. Neither metric indicates significant differences in target-firm pre-deal risk between the categories of private equity deals (sole-sponsored and club) on average,¹⁴ though there is weak evidence that club targets have lower return volatility at the median. The targets of both sole-sponsored and club private equity deals have significantly lower total volatility and higher beta than the targets of acquisitions by publicly traded acquirers do.

As an additional measure of risk, we also examine the target's operating risk (historical cash flow volatility, as measured by the volatility of EBITDA to assets based on up to ten years of history, with a minimum of five). We also examine target complexity (as measured by the number of reported segments in the Compustat segment files) and information asymmetry surrounding the target (as measured by analyst forecast errors in the Institutional Brokers' Estimate System (IBES)). We find no significant differences between club and sole-sponsored private equity target firms along these dimensions. To conserve space, we do not tabulate these results.

Taken together, our univariate comparisons of target characteristics suggest that it is unlikely that the lower gains to club deal target shareholders are due to

¹⁴ As an alternative, we estimated summed betas, where we add up the betas with respect to today's and yesterday's excess market returns. We found no significant difference in these betas between club and sole-sponsor targets. We thank an anonymous referee for this suggestion.

differences in pre-deal target performance, to the premium being incorporated in the stock price before deal announcement, or to differences in risk.

In unreported analysis, we also estimate probit, linear probability, and logit models for club formation using the combined sample of sole-sponsored and club private equity deals. In these models the dependent variable is an indicator variable for club deals and the independent variables are the target characteristics in Table 4 plus year fixed effects. The results indicate that clubs do involve larger targets on average, but the coefficients on the other variables in Table 4 are not significant. This result is consistent with the conclusions from Table 4 with the exception that the difference in institutional ownership is no longer significant.

4.4. Target return regressions

4.4.1. Full-sample analysis

To further investigate the possibility that the differences in gains to target shareholders between club deals and other transactions may be due to differences in observable target or deal characteristics, we estimate regressions in which we use three of the return measures in Table 3 as the dependent variables: announcement returns (CAR3), buy-and-hold abnormal markups, and buy-and-hold abnormal premiums. Because the differences in runups between sole-sponsored and club LBOs are insignificant in Table 3, and because runups are inherently noisy, the total deal premium (which is the sum of the runup and markup) may be too noisy to capture salient differences between sole-sponsored and club private equity deals. This is the reason for reporting regressions with buy-and-hold abnormal markups.¹⁵

Our main indicator variables of interest are whether the acquirer is a club, a private firm, or a public firm. Thus, the omitted category is sole-sponsored private equity firms. The other explanatory/control variables are observable target and deal characteristics, and are all relatively standard in the M&A literature. One of the control variables is the natural logarithm of target size, for consistency with the prior literature, but our results are robust to additional linear and quadratic size controls. Most of our control variables are defined in the prior section. Those that are not previously defined are: indicator variables for the method of payment (cash or a mix); the bidder's toehold at announcement; an indicator variable for whether the offer is a tender offer; an indicator variable for whether target managers have a hostile reaction to the offer; and an indicator variable for whether there are other bidders competing to acquire the target firm (Pre-competition: equal to one if there is another offer in the SDC database for the same target in the six months prior to the bid announcement date). These variables are all defined using SDC data. We also include year and industry (based on two-digit SIC codes) fixed effects to capture as much remaining unmodeled hetero-

geneity as possible, cluster standard errors by year, and correct the standard errors for heteroskedasticity.¹⁶

The results are displayed in Panel A of Table 5. The results in the first three columns represent our base specifications. In these regressions the coefficient on the club indicator variable is significantly negative in the announcement return and buy-and-hold abnormal markup regressions. These specifications suggest that target shareholders receive 6.8% and 8.6%, respectively, less as a percentage of pre-bid equity value in club deals compared to sole-sponsored LBOs. The club deal coefficient in the buy-and-hold abnormal premium regression is also negative but, because runups are noisy but do not differ between sole-sponsored and club LBOs, insignificantly different from zero.¹⁷

We hypothesize that target firms with high institutional ownership will be less likely to accept a low offer price, due to the strong incentives and sophistication of their institutional owners to negotiate a fair price at the bargaining table. To the extent that clubs inadvertently or intentionally constrain competition, we expect institutional owners to mitigate the pricing consequences. On the other hand, since the assumption maintained throughout our analysis is that sole-sponsored LBOs are competitive even if club deals are not, we do not expect to find a similar effect in sole-sponsored deals.

We investigate this idea by adding the target firm's institutional ownership (as in Table 4) to the regressions in columns 4, 5, and 6 in Panel A of Table 5. We interact target institutional ownership with our club and sole-sponsored indicator variables to examine whether the effect of institutional ownership differs across different types of acquirers. The first row of these columns shows that the estimated club discount for a target with zero institutional ownership is large and significant—the point estimates range from 14.3 percentage points lower announcement returns to 27.1 percentage points lower abnormal premiums in club deals. The interactions of the club indicator with institutional ownership are all significantly positive, suggesting an incremental effect by which the lower premiums for targets of club deals dissipate as institutional ownership increases, and are completely eliminated for targets with high institutional ownership. Importantly, there is no incremental effect of institutional ownership on target deal-period returns when the acquirer is a sole private equity (PE) firm.

The coefficients on the institutional ownership variable by itself suggest that institutional ownership is associated with higher premiums in all deals, regardless of acquirer type, perhaps because higher quality targets have higher institutional ownership. We emphasize that the incremental club deal effect exists even when

¹⁵ As mentioned earlier, all our results are qualitatively similar if we use the premium defined as SDC-reported final deal price divided by the target's stock price 43 days prior to deal announcement instead of this returns-based measure of premium.

¹⁶ To address the possibility that significance levels may be overstated due to repeat buyers, we also consider clustering by buyer or by both buyer and year and obtain similar significance levels for key coefficient estimates. We thank an anonymous referee for bringing this issue to our attention.

¹⁷ Despite the noisiness of the premium BHAR measure, as discussed below we do find a significant club discount in the pre-2006 period even using premium BHAR.

Table 5

Multivariate regressions explaining target returns. This table contains the results of multivariate regressions of CAR3, Markup BHAR, and Premium BHAR described in Table 3, and expressed in percent. Club, Private, and Public are indicator variables for the deal categories described in Table 1. Pre-2006 is an indicator variable equal to one for deals announced prior to 2006, and zero otherwise. All cash (Mix) is an indicator variable equal to one if the takeover offer is 100% cash (mix of cash and stock), and zero otherwise. Toehold at announcement is the fraction of the target's shares owned by the acquirer at announcement. Tender (Hostile) is an indicator variable equal to one if the takeover offer is a tender offer (receives a hostile reaction from target managers, as defined by SDC), and zero otherwise. Pre-competition is an indicator variable equal to one if another potential acquirer has bid for the target within the six months prior to the bid announcement date, and zero otherwise. PE firm(s) AUM is the total assets under management of all prominent PE firms involved in the deal, measured as total fundraising in the prior ten years, in billions of dollars; the variable is zero for non-PE deals. PE inflow is the total fundraising in the year prior to the deal by all PE funds with a U.S. geographical focus, in billions of dollars. All other explanatory variables are defined in prior tables. All regressions include industry (two-digit SIC code) fixed effects. All regressions in Panel A also include year fixed effects. Standard errors are clustered by year and heteroskedasticity-consistent. *t*-Statistics are in brackets. ***, **, and * indicate that the coefficient estimate is significantly different from zero at the 1%, 5%, and 10% levels, respectively.

	CAR3	Markup BHAR	Premium BHAR	CAR3	Markup BHAR	Premium BHAR	CAR3	Markup BHAR	Premium BHAR
<i>Panel A: Regressions including year fixed effects</i>									
Club	−6.77** [−2.61]	−8.58** [−2.52]	−5.08 [−1.00]	−14.27** [−2.42]	−23.04** [−2.70]	−27.14* [−2.06]	1.30 [0.86]	−0.36 [−0.13]	6.03 [1.00]
Club × Inst. ownership				12.02** [2.20]	28.91** [2.62]	50.14** [2.56]			
Sole PE × Inst. ownership				0.12 [0.02]	4.84 [0.83]	11.58 [1.41]			
Inst. ownership				4.10** [2.48]	7.10* [2.04]	8.65** [2.39]			
Club × Pre-2006							−8.49*** [−4.09]	−12.13** [−2.77]	−23.91*** [−3.16]
Sole PE × Pre-2006							4.00 [1.52]	1.13 [0.31]	−4.84 [−0.84]
Private	−0.93 [−0.47]	0.46 [0.19]	0.84 [0.24]	−0.92 [−0.19]	5.02 [1.02]	10.75 [1.70]	1.99* [1.87]	1.22 [0.70]	−2.90 [−0.88]
Public	2.22 [1.06]	4.06 [1.56]	7.83** [2.23]	2.03 [0.43]	8.06 [1.69]	17.29** [2.71]	5.15*** [4.76]	4.85*** [3.09]	4.14 [1.16]
Ln (Size)	−2.49*** [−6.70]	−3.05*** [−6.69]	−5.48*** [−6.63]	−2.90*** [−6.41]	−3.68*** [−6.78]	−6.51*** [−6.49]	−2.53*** [−6.87]	−3.11*** [−6.84]	−5.60*** [−6.74]
Prior 12-month BHAR	−0.02*** [−3.67]	−0.02 [−0.90]	−0.03 [−0.91]	−0.02*** [−3.08]	−0.02 [−0.84]	−0.03 [−0.84]	−0.02*** [−3.66]	−0.02 [−0.89]	−0.03 [−0.89]
Runup BHAR		−0.21*** [−5.01]		−0.21***		−0.21***			
Ind.-adjusted EBITDA/assets	0.35 [0.10]	0.00 [0.00]	−6.68 [−0.71]	−0.38 [−0.11]	−0.48 [−0.06]	−5.41 [−0.56]	0.35 [0.11]	0.01 [0.00]	−6.63 [−0.70]
Industry-adjusted Q	−0.48** [−2.41]	−0.82*** [−3.10]	−0.59 [−1.10]	−0.43** [−2.14]	−0.76*** [−2.89]	−0.41 [−0.75]	−0.48** [−2.44]	−0.83*** [−3.11]	−0.61 [−1.12]
Beta	2.06** [2.39]	0.34 [0.27]	0.49 [0.21]	2.05** [2.42]	0.11 [0.08]	0.75 [0.31]	2.12** [2.48]	0.44 [0.35]	0.69 [0.29]

Prior 12-month ret. volatility	−0.29 [−0.60]	−0.68 [−0.60]	−0.00 [−0.00]	−0.30 [−0.59]	−0.64 [−0.51]	−0.17 [−0.10]	−0.29 [−0.61]	−0.69 [−0.61]	−0.02 [−0.01]
All cash	4.61*** [3.50]	−0.74 [−0.28]	2.04 [0.59]	4.38*** [2.95]	−1.14 [−0.40]	1.19 [0.33]	4.66*** [3.53]	−0.66 [−0.25]	2.21 [0.65]
Mix	0.82 [0.81]	−2.90* [−1.81]	−2.68 [−1.01]	0.99 [0.88]	−2.52 [−1.54]	−1.79 [−0.67]	0.83 [0.83]	−2.85* [−1.78]	−2.55 [−0.97]
Toehold at announcement	−6.77 [−1.28]	3.21 [0.34]	−20.48** [−2.51]	−6.17 [−1.00]	6.64 [0.60]	−15.97 [−1.69]	−6.60 [−1.26]	3.43 [0.36]	−20.03** [−2.48]
Tender	8.47*** [6.49]	6.12*** [5.62]	7.36*** [5.88]	8.66*** [6.75]	6.60*** [5.94]	8.02*** [6.18]	8.40*** [6.43]	6.05*** [5.55]	7.25*** [5.71]
Hostile	5.20*** [3.10]	10.47*** [4.39]	7.00*** [2.81]	6.05*** [3.59]	11.16*** [4.25]	5.88* [1.87]	5.29*** [3.15]	10.56*** [4.41]	7.13*** [2.87]
Pre-competition	−11.71*** [−8.25]	−9.05*** [−3.78]	2.99 [0.95]	−12.44*** [−8.49]	−9.86*** [−3.73]	3.55 [0.99]	−11.64*** [−8.22]	−8.97*** [−3.76]	3.10 [0.98]
Constant	22.37 [1.67]	7.59 [0.88]	20.41*** [3.44]	18.00 [1.31]	6.96 [0.63]	8.76 [0.81]	19.34 [1.43]	6.78 [0.81]	24.19*** [4.27]
Observations	3382	3382	3382	3171	3171	3171	3382	3382	3382
R-squared	0.15	0.12	0.14	0.15	0.12	0.14	0.16	0.12	0.14

Panel B: Regressions including controls for PE firm(s) assets under management (AUM) and PE industry inflow

	CAR3	Markup BHAR	Premium BHAR	CAR3	Markup BHAR	Premium BHAR	CAR3	Markup BHAR	Premium BHAR
Club	−10.52*** [−4.80]	−13.53*** [−4.28]	−17.00*** [−3.02]	−15.02** [−2.53]	−25.95*** [−3.37]	−33.13*** [−3.04]	4.77 [1.12]	5.85 [0.76]	7.41 [0.67]
Club × PE inflow	0.06 [1.64]	0.07 [1.50]	0.16** [2.37]	0.04 [1.05]	0.01 [0.11]	0.05 [0.60]	−0.03 [−0.81]	−0.05 [−0.66]	−0.01 [−0.06]
PE inflow	0.04* [1.94]	0.07 [1.65]	0.05 [1.05]	0.03 [1.51]	0.05 [1.30]	0.04 [0.72]	0.08** [2.62]	0.16** [2.78]	0.20** [2.79]
Club × Inst. ownership				6.45 [1.08]	24.82** [2.50]	39.57** [2.55]			
Sole PE × Inst. ownership				−1.45 [−0.22]	0.01 [0.00]	3.75 [0.57]			
Inst. ownership				4.06** [2.41]	7.45** [2.16]	8.68** [2.36]			
Club × Pre-2006							−10.29*** [−3.71]	−15.49** [−2.38]	−23.89** [−2.27]
Sole PE × Pre-2006							5.35* [1.80]	3.97 [1.03]	−0.85 [−0.13]
Pre-2006							5.24 [1.47]	13.30* [1.96]	22.00** [2.57]
Assets under management	0.05 [0.45]	0.06 [0.49]	0.14 [1.05]	0.07 [0.59]	0.13 [1.01]	0.27* [2.07]	0.08 [0.68]	0.09 [0.65]	0.14 [0.93]
Private	0.39 [0.18]	2.67 [0.98]	3.93 [0.96]	−0.44 [−0.09]	4.65 [0.89]	10.03 [1.60]	4.35** [2.22]	5.26** [2.17]	2.42 [0.50]
Public	3.29 [1.62]	5.40* [1.95]	9.28** [2.21]	2.37 [0.50]	7.01 [1.42]	15.03** [2.48]	7.26*** [4.30]	8.03*** [3.68]	7.83 [1.50]
Ln (size)	−2.13*** [−4.60]	−2.16*** [−3.38]	−4.35*** [−4.39]	−2.52*** [−4.48]	−2.82*** [−3.78]	−5.39*** [−4.84]	−2.28*** [−4.87]	−2.52*** [−4.04]	−4.92*** [−4.81]
Prior 12-month BHAR	−0.02*** [−4.04]	−0.02 [−0.99]	−0.03 [−1.32]	−0.02*** [−3.48]	−0.02 [−0.93]	−0.03 [−1.29]	−0.02*** [−3.73]	−0.02 [−0.93]	−0.03 [−1.20]
Runup BHAR		−0.19*** [−4.78]		−0.19*** [−4.73]		−0.19*** [−4.73]			

Table 5 (continued)

Panel B: Regressions including controls for PE firm(s) assets under management (AUM) and PE industry inflow									
	CAR3	Markup BHAR	Premium BHAR	CAR3	Markup BHAR	Premium BHAR	CAR3	Markup BHAR	Premium BHAR
Ind.-adjusted EBITDA/Assets	2.65 [0.73]	6.09 [0.70]	2.39 [0.25]	1.79 [0.47]	5.37 [0.58]	3.65 [0.37]	2.15 [0.61]	4.87 [0.59]	0.44 [0.05]
Industry-adjusted Q	−0.52** [−2.55]	−0.88*** [−3.49]	−0.59 [−1.16]	−0.45** [−2.11]	−0.77*** [−3.00]	−0.34 [−0.66]	−0.51** [−2.50]	−0.86*** [−3.54]	−0.55 [−1.09]
Beta	0.85 [0.75]	−2.33 [−1.19]	−3.56 [−1.20]	0.77 [0.72]	−2.56 [−1.39]	−3.32 [−1.16]	1.37 [1.27]	−1.04 [−0.63]	−1.46 [−0.57]
Prior 12-month ret. volatility	0.63 [1.19]	1.58 [1.53]	3.36** [2.39]	0.55 [0.99]	1.47 [1.28]	3.07* [2.00]	0.34 [0.64]	0.86 [0.86]	2.17 [1.53]
All cash	4.46*** [3.61]	−1.02 [−0.38]	1.03 [0.30]	4.23*** [3.06]	−1.33 [−0.46]	0.14 [0.04]	4.67*** [3.92]	−0.45 [−0.17]	2.00 [0.60]
Mix	1.11 [1.44]	−2.32 [−1.37]	−2.58 [−1.00]	1.13 [1.34]	−2.17 [−1.28]	−2.01 [−0.79]	1.08 [1.38]	−2.32 [−1.35]	−2.52 [−0.99]
Toehold at announcement	−6.81 [−1.27]	3.33 [0.36]	−19.78** [−2.48]	−5.16 [−0.84]	9.39 [0.91]	−12.18 [−1.43]	−6.21 [−1.18]	4.58 [0.50]	−17.64** [−2.25]
Tender	8.62*** [6.36]	6.64*** [5.12]	9.43*** [6.26]	9.04*** [6.93]	7.35*** [5.88]	10.41*** [6.74]	8.46*** [6.36]	6.28*** [5.05]	8.85*** [5.63]
Hostile	5.16*** [2.94]	10.85*** [4.25]	7.73*** [2.88]	6.02*** [3.36]	11.69*** [4.16]	6.87* [2.04]	5.20*** [2.95]	10.85*** [4.16]	7.70** [2.71]
Pre-competition	−12.03*** [−8.20]	−9.68*** [−3.69]	3.36 [0.92]	−12.43*** [−7.88]	−9.99*** [−3.46]	4.43 [1.13]	−11.84*** [−8.22]	−9.20*** [−3.67]	3.99 [1.12]
Constant	18.47 [1.44]	1.29 [0.22]	14.27** [2.35]	17.79 [1.30]	−4.05 [−0.51]	3.93 [0.54]	9.36 [0.84]	−14.06* [−1.99]	−5.28 [−0.46]
Observations	3382	3382	3382	3171	3171	3171	3382	3382	3382
R-squared	0.14	0.09	0.11	0.14	0.09	0.10	0.14	0.10	0.11

controlling for any aspect of target quality that may be proxied by institutional ownership.¹⁸

In the last three columns of Panel A of Table 5 we interact our club and sole-PE indicator variables with an indicator variable for deals announced prior to 2006, the year in which the financial media turned a spotlight onto the practice of club deals by private equity firms and the U.S. Department of Justice started its informal inquiry into the practice.¹⁹ It is possible that the attention placed on club deals by the media and the U.S. government caused private equity firms to be more timid in exploiting the lack of competition created by private equity clubs and, therefore, less aggressive with deal pricing in 2006 and 2007. It is also possible that increased funding to private equity firms in 2006 and 2007 increased competition for target firms such that the reduction in competition associated with clubs had negligible pricing consequences in those years.

Consistent with these possibilities, the coefficients on the interactions of the pre-2006 and club indicator variables are significantly negative for all measures of returns (announcement, markup, and premium), while the coefficients on the club indicator variable by itself are insignificantly different from zero. The net pre-2006 club discount, represented by the linear expression (Club + Club $\times$ Pre-2006 – Sole PE $\times$ Pre-2006), is significantly negative for all three return measures, and ranges from 11.2 to 13.6 percentage points of pre-bid target equity value in magnitude. Therefore, essentially all the club deal discount that we observe is driven by club deals prior to 2006. The same pattern is not evident in the coefficients on the interactions of the sole-PE indicator variable.

All of the results discussed above are also robust to a variety of additional or alternative controls. Specifically, the results are robust to scaling EBITDA by market value 43 days before announcement rather than assets; to a control for return on invested capital, measured by EBITDA scaled by total deal value; to a control for firm size based on assets; to controls for governance variables (where data are available), specifically the fraction of independent directors on the target's board and the Gompers, Ishii, and Metrick (2003) index; to a control for the number of segments in the firm (where data are available); and to a control for the age (a measure of experience) of the private equity firm(s).

The results are also robust to including private equity firm fixed effects in the regressions. This suggests that our results are not driven by the ability of some private equity partnerships to pay low prices whether acting alone or as part of a club. That said, we do not include these fixed

effects in our main specifications because including them may not produce a well-specified analysis of the potential club discount. To illustrate the point, suppose that, holding all else constant, two private equity firms are both able to pay 10% less when acting alone, but 15% less when they form a club together. This is a true club discount, but including private equity firm fixed effects would result in an estimated 5% ($= -15\% - (-10\% - 10\%)$) club premium.²⁰

The coefficients on the control variables in Panel A of Table 5 generally have signs and significance levels that are consistent with the prior literature. The coefficients on the pre-bid competition indicator variables appear unusual, because these coefficients are significantly negative (indicating discounts of about 10 percentage points) in most of the specifications in the table, even though intuition suggests that pre-bid competition would be associated with higher premiums. However, it is likely that this coefficient reflects the fact that prior bids reduce the surprise associated with the current bid. The same result is reported in Officer (2003). There is also evidence of markup pricing (Schwert, 1996) in our regressions. Specifically, the coefficients on the runup variables in the markup regressions are all significantly negative and equal to about 0.2 in magnitude, suggesting that acquirers only shade their bids downwards by about 20% of the pre-bid runup.

In Panel B of Table 5 we repeat the analysis in Panel A including controls for private equity firm assets under management (AUM) and private equity fund inflows calculated using data from Preqin. This analysis is intended to provide insight into whether capital constraints are likely to drive the pricing of club deals, and whether increased inflows to PE funds around 2006, corresponding to looser capital constraints and more competition for deals, can explain the structural break in club deal pricing around 2006.²¹ We measure the AUM of the prominent PE firms involved in a deal as their total fundraising in the ten years prior to the deal (i.e., for club deals, we add the AUMs of the individual prominent PE firms; the variable is zero for non-PE deals). We measure the annual inflow to the PE industry as the total fundraising in the year prior to the deal by all PE funds with a U.S. geographical focus. In Panel B of Table 5, we drop year fixed effects because they would subsume this inflow variable.

The results in Panel B of Table 5 continue to support our conclusions from Panel A. The estimated magnitude of the club deal level effect is stronger, the cross-sectional institutional ownership effect is slightly weaker, and the time-series structural break around 2006 is just about the

¹⁸ These results are qualitatively unaffected if we instead use a measure of *long-term* institutional ownership, defined as the percentage ownership of institutions who also held an ownership stake four quarters prior (similar to the rationale in Gaspar, Massa, and Matos, 2005). One caveat to our analysis of institutional ownership is that hedge funds and other blockholders may not report holdings as institutions. Because of this, our institutional ownership measure is an imperfect measure of the holdings of shareholders with strong incentives and sophistication to maximize value.

¹⁹ The direct effect of the pre-2006 indicator variable is subsumed by the year fixed effects.

²⁰ A potential solution is to include “club” fixed effects—in the case of a sole-sponsored LBO, these would be private equity firm fixed effects. However, a subset of these fixed effects would absorb the club indicator variable, a key variable of interest in our regressions. Moreover, the structural break around 2006 would be identified off a few clubs that transact both before and after 2006. Likewise, the cross-sectional institutional ownership effect would be identified off clubs that deal with different levels of institutional ownership in separate transactions. None of our conclusions are affected by including such fixed effects. We thank an anonymous referee for bringing this solution to our attention.

²¹ We thank an anonymous referee for suggesting this analysis.

same. The coefficient estimate on AUM is insignificant in eight of the nine specifications. The positive and sometimes significant coefficient estimates on the PE inflow level variable are consistent with the “money chasing deals” phenomenon described in [Gompers and Lerner \(2000\)](#). However, such money chasing deals do not explain the 2006 structural break in the club deal discount. The interaction of the inflow measure with the club indicator variable is insignificant in all nine specifications except one. Crucially, our finding of a structural break in 2006 (the last three specifications) continues to be significant.

4.4.2. Private equity sample analysis

Our analysis in [Table 5](#) includes M&A transactions completed by non-private equity bidders. The advantage

of including these transactions is that they serve as a control sample that allows us to estimate the effects of a battery of control variables. However, it may be that firms targeted by prominent private equity bidders differ in important unobservable ways from those targeted by other types of bidders.

Therefore, in [Table 6](#) we confirm that the conclusions from [Table 5](#) continue to hold in analysis using only the sample of 201 private equity transactions. Panel A repeats the regression analysis in Panel A of [Table 5](#), but we drop some of the control variables given the limited number of observations. We retain controls for target size and risk, as well as year fixed effects.

The first three columns of Panel A of [Table 6](#) confirm that club deals do involve significantly smaller gains to

Table 6

Prominent private equity bidder sample analysis This table contains the results of regression and matching analyses for the sample of 201 private equity transactions conducted by prominent private equity firms. The dependent variables, CAR3, Markup BHAR, and Premium BHAR, are described in [Table 3](#), and expressed in percent. All explanatory variables are defined in prior tables. All regressions in Panel A contain year fixed effects, and standard errors are clustered by year and heteroskedasticity-consistent. The first column of Panel B reports club treatment effects from kernel propensity-score matching using propensity scores from a probit model for club formation with the control variables in Panel A and year fixed effects. The second column of Panel B reports treatment effects based on [Abadie-Imbens \(2007\)](#) matching estimator with matching conducted based on the control variables in Panel A. *t*-Statistics are in brackets. ***, **, and * indicate that the coefficient estimate is significantly different from zero at the 1%, 5%, and 10% levels, respectively. In the last row of Panel A, the *p*-values are two-tailed values.

Panel A: Multivariate regressions explaining target returns									
	CAR3	Markup BHAR	Premium BHAR	CAR3	Markup BHAR	Premium BHAR	CAR3	Markup BHAR	Premium BHAR
Club	−7.37*	−9.37**	−11.65**	−16.49**	−28.56**	−39.85**	2.97*	0.52	−2.31
	[−2.07]	[−2.45]	[−2.11]	[−2.31]	[−2.29]	[−2.44]	[1.95]	[0.24]	[−0.36]
Club*Inst. ownership				14.91	29.04*	43.19*			
				[1.66]	[1.94]	[2.02]			
Inst. ownership				6.16	7.58	10.31			
				[0.72]	[0.71]	[0.93]			
Club × Pre-2006							−16.03***	−15.33***	−14.47
							[−3.67]	[−3.35]	[−1.54]
Ln (size)	−1.34	−2.17	−1.62	−1.68	−2.35	−1.67	−2.12*	−2.92*	−2.33
	[−1.37]	[−1.39]	[−1.15]	[−1.42]	[−1.12]	[−0.84]	[−1.75]	[−1.73]	[−1.63]
Prior 12-month BHAR	−0.05*	−0.00	−0.06	−0.05	−0.00	−0.05	−0.04	0.01	−0.05
	[−1.83]	[−0.04]	[−0.75]	[−1.31]	[−0.03]	[−0.79]	[−1.44]	[0.16]	[−0.63]
Runup BHAR		−0.20			−0.23*			−0.20	
		[−1.43]			[−1.82]			[−1.52]	
Beta	−0.11	2.24	−2.93	−0.55	1.08	−5.56	1.92	4.18	−1.10
	[−0.03]	[0.41]	[−0.45]	[−0.11]	[0.18]	[−0.83]	[0.46]	[0.80]	[−0.17]
Prior 12-month ret. volatility	2.57	0.92	0.11	4.32	2.79	1.94	2.51	0.86	0.06
	[1.03]	[0.22]	[0.02]	[1.20]	[0.57]	[0.32]	[0.98]	[0.20]	[0.01]
Constant	11.68***	13.07*	28.72**	4.30	5.42	20.50	9.42**	10.93	26.67**
	[3.17]	[1.95]	[2.73]	[0.61]	[0.57]	[1.37]	[2.33]	[1.57]	[2.49]
Observations	198	198	198	169	169	169	198	198	198
R-squared	0.26	0.29	0.27	0.32	0.37	0.34	0.31	0.32	0.28
Club+Club × Pre-2006							−13.05	−14.82	−16.78
<i>p</i> -Value							0.004	0.006	0.032
Panel B: Estimated treatment effect from matching analysis for club effects in target returns									
	Propensity-score matching			Abadie-Imbens matching					
CAR 3			−9.75***			−6.03**			
			[−3.05]			[−2.40]			
Markup BHAR			−12.11***			−7.28**			
			[−2.87]			[−2.10]			
Premium BHAR			−14.74***			−11.86**			
			[−2.74]			[−2.38]			

target shareholders. The estimated club discounts are somewhat larger in magnitude than those in Table 5, ranging from 7.4 to 11.7 percentage points.

The next three columns confirm the result that the club discount is mitigated by high institutional ownership. In all three columns the coefficients on the club indicator are significantly negative. The coefficients on the interaction of institutional ownership with the club indicator are all positive, and significantly so for the buy-and-hold abnormal returns. The magnitudes of these interactions are similar to those in Table 5. The insignificant coefficients on institutional ownership by itself again indicates that, consistent with Table 5, there is no similar effect for sole-sponsored private equity deals.

The final three columns confirm that the club discount is concentrated before 2006 and virtually disappears in 2006 and 2007. In the announcement return and abnormal markup regressions, the coefficients on the interaction of the club and pre-2006 indicator variables are significantly negative, with magnitudes of 16.0 and 15.3 percentage points. In the buy-and-hold abnormal premium regression, this coefficient is negative but not significant at conventional levels. However, the sum of the coefficients on the club indicator and the interaction term ($-2.3 + -14.5 = -16.8$, or a 16.8 percentage point discount) is significantly negative (p -value=0.03). This sum of coefficients represents the club deal discount before 2006 ($\text{Club}=1$ and $\text{Pre-2006}=1$), and is also significantly negative in the prior two columns with p -values of 0.00 and 0.01, respectively.

In Panel B of Table 6, we confirm the robustness of our finding of a club discount when we employ matching, rather than regression, analysis. The first column of Panel B contains estimated club treatment effects on premiums from a propensity-score matching analysis in which propensity scores are determined using a probit model for club formation that uses the control variables in Panel A of Table 6 plus year fixed effects as independent variables. The estimates are computed using a kernel estimator that weights matches by their closeness in propensity scores. The second column of Panel B contains estimated club treatment effects on premiums using the bias-corrected matching estimator of Abadie and Imbens (2007), in which the matching is conducted using the control variables in Panel A of Table 6. The estimated club treatment effects, which essentially represent average differences in premiums between club and sole-sponsored deals with similar observable characteristics, are similar in magnitude to the regression estimates in Table 5 and Panel A of Table 6.

Overall, the evidence presented in this section suggests that target shareholders gain less in club deals than in sole-sponsored private equity deals, that this discount is concentrated before 2006, and that the discount is more pronounced in target firms with low institutional ownership.

5. Multiples-based measures of acquisition prices

While the M&A literature typically focuses on returns-based measures of premiums paid to target shareholders, as we have done so far, the private equity literature

typically focuses on pricing measures based on multiples of an accounting variable, most commonly sales or EBITDA. Unfortunately, multiples are a noisy valuation metric and it is not clear what the appropriate benchmark for comparison should be. Comparisons of multiples also suffer from a potential omitted variables bias. In general, the main advantage of studying valuation multiples is that they facilitate comparison of acquisitions of public and private targets.

In this paper, we do not have to rely on noisy valuation multiples because we study acquisitions of public targets for which stock-return data are available. Indeed, to the extent that the market does not anticipate the deal, post-deal announcement stock returns (i.e., markups) should represent the true wealth gain experienced by target shareholders, since stock prices reflect all available value-relevant information in an efficient market. Nevertheless, valuation multiples can serve as a robustness check for returns-based measures of premiums because the analysis does not require a choice of measurement window and because deal valuation multiples should not be affected even if the target shares are misvalued by the market before the deal. For these reasons, and in keeping with the private equity literature, we analyze transaction multiples in the following way.

We use the “comparable industry transaction method” described in Kaplan and Ruback (1995) and employed in Officer (2007). We measure deal multiples by dividing each of four different measures of enterprise value by annual sales or EBITDA (from Compustat for the fiscal year ending immediately prior to the deal announcement date). The reason for using four potential measures of enterprise value is that each is potentially flawed in its own way.

The four measures of target enterprise value we use are:

- (1) Deal value excluding assumed liabilities (from SDC)+total debt (from Compustat)–cash (from Compustat);
- (2) Market capitalization at the delisting date (from CRSP)+total debt (from Compustat)–cash (from Compustat);
- (3) (Delisting stock price (split adjusted) multiplied by the number of shares outstanding at the beginning of the runup period (both from CRSP))+total debt (from Compustat)–cash (from Compustat);
- (4) (Price paid to acquire each target share (from SDC) multiplied by the number of shares outstanding at the beginning of the runup period (from CRSP))+total debt (from Compustat)–cash (from Compustat).

The principal difference between these four measures is the measurement of the market capitalization of the target firm (including the premium paid by the acquirer). Values (1) and (4) above employ SDC data to calculate this value, while values (2) and (3) employ CRSP data only. Value (2) differs from value (3) if the number of shares outstanding changes between the beginning of the runup period and the delisting date: this could happen in, for

example, two-tier offers (which are likely part of our sample).

Cash (from Compustat for the fiscal year ending immediately prior to the deal announcement date) is subtracted to ensure that we are consistent in capturing payment by the acquirer for the real assets and operations of the target firm: private equity firms, for example, often use the target's cash to directly offset any liabilities assumed as part of the acquisition (Kaplan and Stein, 1993). By subtracting cash we are also ensuring that our results are robust to systematic differences in excess cash holdings by targets of the various types of acquirers. We also discard observations from the sample if either the numerator (enterprise value) or denominator (EBITDA or sales) is negative, and then winsorize each of the multiples independently at the 5th and 95th percentiles.

We then use these deal multiples to produce estimates of the “discount” in private equity sponsored acquisitions (as in Officer, 2007). Specifically, for each private equity sponsored deal in our sample (club or sole-sponsored), we find a matching portfolio of non-private equity sponsored deals and calculate the average multiple for that portfolio. Each portfolio is formed by finding all non-private equity sponsored deals announced in a three-year window centered on or trailing the announcement date of the private equity sponsored deal and for which the target is in the same two-digit SIC code industry as the target of the private equity sponsored deal. We compute the percent difference between the sales or EBITDA multiple for each private equity sponsored deal and the comparable multiple for the matching portfolio. These percent differences are measures of the acquisition discount for each private equity sponsored deal (more negative numbers represent larger discounts). For brevity, we average the discounts across the different definitions of enterprise value to obtain two discount measures for each deal (one for sales multiples and one for EBITDA multiples; we obtain similar results with each of the four different definitions of enterprise value), and compare the resulting discount measures across club and sole-sponsored private equity deals. This analysis is therefore akin to a differences-in-differences methodology.

Table 7 presents regressions with the acquisition discounts for private equity sponsored deals as the dependent variable. The sample includes only private equity sponsored deals (sole-sponsored or club), because the discount is calculated relative to portfolios including all other deals (non-private equity sponsored) in the main sample.

In Panel A of Table 7, the matching window is a three-calendar-year window centered on the announcement date (as in Officer, 2007), and we control for the natural log of target size (market value of equity from CRSP 43 days prior to the announcement) in lieu of matching on size (which is addressed below). The independent variables of interest to us are the club indicator variable and its interaction with the pre-2006 indicator variable. In both specifications (based on sales or EBITDA multiples) the interaction of the club indicator and the pre-2006 indicator has a significantly negative coefficient, consistent with the returns-based results in Tables 5 and 6.

Table 7

Multivariate regressions explaining acquisition discounts based on deal multiples. This table contains the results of multivariate regressions where the dependent variables are acquisition discounts for private equity sponsored deals based on enterprise value multiples, EV/Sales and EV/EBITDA. The acquisition discount is defined as the percent difference between a deal multiple for a private equity sponsored deal and the average multiple for an equal-weighted portfolio of non-private equity sponsored deals. A negative discount means that the private equity deal multiple is lower than that of the comparison portfolio. There are four different measures of enterprise value (EV) employed in the numerators of the multiples, and the results reported in this table are based on the average discount across the four different EV/Sales or EV/EBITDA multiples. In Panel A, each portfolio contains all non-private equity sponsored deals announced in the three year window centered on the announcement date of the private equity sponsored deal and for which the target is in the same two-digit SIC code industry as the target of the private equity sponsored deal. In Panel B, each portfolio contains all non-private equity sponsored deals announced in the three year window preceding the announcement date of the private equity sponsored deal, for which the target is in the same two-digit SIC code industry as the target of the private equity sponsored deal, and for which the target is within +/- 50% of the market value of the target of the private equity sponsored deal, measured as market value 43 trading days before takeover announcement. All explanatory variables are defined in prior tables. Standard errors are clustered by year and heteroskedasticity-consistent. *t*-Statistics are in brackets. ***, **, and * indicate that the coefficient estimate is significantly different from zero at the 1%, 5%, and 10% levels, respectively. In the last row of Panels A and B, the *p*-values are two-tailed values.

Panel A: Three-year match window centered on deal announcement		
	EV/sales	EV/EBITDA
Club	24.44*** [4.31]	14.95 [1.62]
Club × Pre-2006	−51.91*** [−4.11]	−26.95* [−2.06]
Pre-2006	13.94 [1.33]	13.13 [1.00]
Ln (size)	13.49*** [3.86]	0.78 [0.20]
Constant	−4.82 [−1.40]	−23.51** [−2.23]
Observations	183	166
R-squared	0.08	0.02
Club+Club × Pre-2006	−27.47	−12.01
<i>p</i> -Value	0.017	0.080
Panel B: Three-year match window preceding deal announcement and size matched		
	EV/sales	EV/EBITDA
Club	62.61*** [9.41]	31.05** [2.69]
Club × Pre-2006	−97.16*** [−5.57]	−44.74*** [−3.25]
Pre-2006	29.58 [1.50]	7.04 [0.36]
Constant	−7.67 [−0.58]	−14.33 [−0.82]
Observations	128	108
R-squared	0.04	0.05
Club+Club × Pre-2006	−34.55	−13.68
<i>p</i> -Value	0.045	0.085

Furthermore, the net pre-2006 club discount (represented by (Club+Club × Pre-2006), reported at the bottom of the table) is significantly negative for both sales- and EBITDA-

based discount measures. The estimated magnitude of the net pre-2006 club discount using EBITDA-based discount measures, 12.0%, is similar to the estimates using target stock returns in Section 4, while that using sales-based multiples, 27.5%, is considerably larger.

While matching comparable non-private equity transactions only by industry (and controlling for size in the regression) has the benefit of producing more matches than implementing an additional match on size would, one might be concerned that a control for (log) size may not adequately capture the relation between target size and acquisition multiples. Moreover, one might be concerned that the three-year centered window incorporates information that was not available to the private equity firms, or their targets, at the time of the deal (i.e., information about future deals). To address these concerns, Panel B of Table 7 employs matches from the three-calendar-year window immediately preceding the announcement date and we further match on size (and so omit a control for size in the regressions); we require that matching non-private equity targets have a market value within $\pm 50\%$ of the market value of the private equity target, measured as the market value 43 trading days before the takeover announcement.

As in Panel A, the specifications in Panel B show a significantly negative coefficient on the interaction of the club and the pre-2006 indicator variables, and the net pre-2006 club discount (sum of the coefficients on the club indicator variable and the interaction term) is again significantly negative. The estimated magnitudes of the net pre-2006 club discount are similar to those in Panel A. Overall, the results in Table 7 are consistent with the evidence in Tables 5 and 6 that there is a club deal discount in the pre-2006 period.²²

One common facet of multiples-based analyses is extreme noise in the data (see, for example, Officer, 2007). While we confirm our principal result from Table 5, that club deals are completed at significantly lower prices than sole-sponsored deals are before 2006, the noise in the multiples-based measures is obvious in the coefficients and low *R*-squared values reported in Table 7. We find no evidence of club premiums after 2006 using returns-based measures of premiums in Table 5, and yet there is some evidence of such in Table 7 (the coefficient on the club indicator by itself). Moreover, some of the statistical significance of the effects in Table 7 is weaker than in Table 5. Both of these conditions are likely the direct result of noise coming from acquisition multiples.

6. Evidence on post-announcement competition and benign rationales for club formation

6.1. Post-announcement competition

To further investigate the possibility that our findings of lower pricing in club deals are due to club formation reducing competition for target firms, we examine the extent to which the target firm receives binding competing bids *after* announcement of the private equity takeover, which we term post-announcement bid competition. This measure of competition contrasts with the pre-announcement bid competition (the extent to which the target firm receives competing expressions of interest or binding bids *before* announcement of the private equity takeover), which is the focus of Boone and Mulherin (2009). Given the possibility of collusion, non-binding or informal expressions of interest could be shams, and it is even possible that purposefully low binding bids are orchestrated prior to the announcement of the deal to provide the appearance of competition.²³ In contrast, post-announcement competing bids truly do compete with the bid of the private equity firm(s) involved in the deal.

In Table 8, we present evidence on the incidence of post-announcement competition, and test whether club deals encounter less post-announcement competition than sole-sponsored private equity deals. Our approach relies on publicly disclosed bids in SDC supplemented with a Factiva search for news articles related to competing bids for any of the private equity targets in our sample. We construct an indicator variable for post-announcement competition that is equal to one if either SDC or a Factiva search indicates at least one competing bid by another potential acquirer in the six-month period following the announcement date in SDC, and zero otherwise.

Panel A of Table 8 shows post-announcement competition in 20 out of 131 sole private equity sponsored deals (15.3%) and four out of 70 club deals (5.7%), consistent with club deals depressing bid competition. Interestingly, in only four of these 24 cases is the competing bidder another prominent private equity firm. The difference is statistically significant in a univariate probit regression (column 1 of Panel B). The average target return from the initial bid to the final bid in the 20 sole private equity sponsored deals with post-announcement competition is 11.5% whereas the corresponding average for the four club deals with post-announcement competition is just 5.2%.

We consider separately a subsample in which we expect the potential anticompetitive effects of clubs to be

²² The results in Table 7 continue to hold if we adopt a different form of matching which relies on Standard & Poor's (S&P)-500-level-adjusted deal values. Specifically, instead of including a specific match on time, we adjust the equity values of the comparable transactions by multiplying them by the ratio of the S&P-500 index level at the private equity deal announcement date to the S&P-500 index level at the comparable deal announcement date. We then match using either industry alone (comparable to the results in Panel A) or industry and pre-deal target market value of equity (comparable to the results in Panel B), and find qualitatively similar results to those reported in Table 7. We thank an anonymous referee for this suggestion.

²³ Even in the absence of repeat play among a handful of private equity firms, bids in a one-shot auction may not reliably measure the extent of real competition because it is not uncommon for private equity firms to submit separate bids in an auction, and then for the eventual winner to later invite losers to participate in the same deal. In one such case, KKR won the auction for PanAmSat in April 2004, beating out the Carlyle Group and Providence Equity Partners, only to include them in the deal two months later. "Carlyle, Providence Will Each Buy 27% PanAmSat Stakes," *Bloomberg*, June 29, 2004.

Table 8

Post-announcement competition. This table reports evidence on the incidence of post-announcement competition in leveraged buyouts sponsored by prominent private equity firms. Post-announcement competition is an indicator variable equal to one if another potential acquirer bids for the target in the six months after the deal announcement date (gathered through SDC and Factiva news searches), and zero otherwise. All of the explanatory variables are defined in prior tables. Top 5 refers to the PEI list and includes Carlyle, KKR, Goldman, Blackstone, and TPG. The coefficients reported in Panel B are marginal responses. Standard errors are heteroskedasticity-consistent and *t*-statistics are reported in brackets. ***, **, and * indicate that the coefficient estimate is significantly different from zero at the 1%, 5%, and 10% levels, respectively.

Panel A: Frequency distributions						
	Sole PE	%	Club	%	All	%
Full sample						
Competing bid	20	15.3	4	5.7	24	11.9
No. of competing bid	111	84.7	66	94.3	177	88.1
N	131		70		201	
Top 5						
Competing bid	9	22.5	1	2.5	10	12.5
No. of competing bid	31	77.5	39	97.5	70	87.5
N	40		40		80	
Panel B: Probit regressions explaining post-announcement competition						
Sample	All (1)		PEI5 (2)		All (3)	
Club	−0.10** [−2.00]		−0.20** [−2.51]		−0.33*** [−2.97]	
Club × Institutional ownership					0.32** [2.49]	
Institutional ownership					−0.04 [−0.62]	
Ln (size)					0.00 [−0.34]	
Prior 12-month return					−0.00* [−1.70]	
CAR3					−0.00** [−2.21]	
Toehold at announcement					−0.40 [−0.80]	
Tender					0.07* [1.81]	
Hostile					0.34*** [2.62]	
Observations	201		80		169	
Pseudo R-squared	0.03		0.14		0.21	

strongest, namely clubs involving the top-five private equity firms from the PEI50 rankings, which are Carlyle, KKR, Goldman Sachs, Blackstone, and TPG. Our sample has 80 deals involving these private equity firms, 40 of which are sole-sponsored and 40 of which are clubs. Panel A of Table 8 shows post-announcement competition in nine out of the 40 sole-sponsored deals (22.5%) but only one of the 40 club deals (2.5%). This difference is statistically significant (column 2 of Panel B).

Our main evidence on prices paid in club deals suggests that sophisticated institutional investors play an important role in mitigating the inadvertent or intentional anticompetitive effect of clubs. Column 3 of Panel B shows that the negative relation between club deals and post-announcement competition is mitigated by high institutional ownership, which further suggests that encouraging or soliciting post-announcement bid competition is one of the ways in which institutional investors improve prices. The insignificant coefficient on

institutional ownership by itself indicates that, similar to our pricing results, there is no similar effect for sole-sponsored private equity deals.

Overall, the evidence on post-announcement competition is consistent with our pricing results and with the view that club deals are associated with reduced competition for target firms.

6.2. Financing commitments

To further study potential differences between club and sole-sponsored private equity deals in our sample, we read definitive merger proxy statements (DEFM14s, or in the few cases of tender offers, SC TOs) and do not detect important differences in deal characteristics that may explain our finding of low prices in club deals. In the process, we collect information on (i) debt and equity commitments made to target shareholders, and (ii) total funds needed to complete

Table 9

Financing commitments and debt financing terms. This table reports evidence on financing commitments and debt financing terms in leveraged buyouts sponsored by prominent private equity firms. Information on debt and equity commitments made by banks and private equity firms, respectively, for deals completed in 1994–2007, is gathered from definitive merger proxy or tender offer statements filed with the SEC. Data on debt financing terms are gathered from DealScan. For deals with multiple credit facilities in DealScan, data are weighted using face values. No. of lenders is the number of lenders providing debt financing for the deal. Spread is the interest rate spread over LIBOR, and Adj. spread is the spread minus the average spread on all LBO debt taken out in the previous calendar quarter, both expressed in basis points per year. Max. debt/EBITDA is the maximum allowed debt-to-EBITDA ratio under the terms of the credit facility, and Adj. max. debt/EBITDA is Max. debt/EBITDA relative to the average corresponding covenant on all LBO debt taken out in the previous calendar quarter. Post-LBO leverage is the ratio of debt-to-debt plus equity under the capital structure immediately post-LBO. All other variables are defined in previous tables. In Panel C, the reported numbers are means for the samples of Sole PE and Club deals. Standard errors are heteroskedasticity-consistent and *t*-statistics are reported in brackets. ***, **, and * indicate that the coefficient estimate is significantly different from zero at the 1%, 5%, and 10% levels, respectively.

Panel A: Total financing commitments relative to total funds needed					
	Sole PE	Club	All		
Mean (%)	99.4	101.6	100.4		
Median (%)	100.0	101.2	100.0		
Under	19	15	34		
Over	15	5	20		
Equal	27	25	52		
N	61	45	106		
Panel B: Equity commitments relative to total financing commitments					
	Sole PE	Club	All		
Mean (%)	33.7	38.2	35.6		
Median (%)	30.8	34.6	31.9		
N	69	52	121		
Panel C: Debt financing terms					
	No. of lenders	Spread	Adj. spread	Max. debt/EBITDA	Adj. max debt/EBITDA
Sole PE (mean)	7.5	289	– 32	6.1	– 0.2
N	61	58	58	28	28
Club (mean)	10.5	286	– 43	7.0	0.4
N	46	46	44	24	23
Panel D: Regressions explaining debt financing terms					
	No. of lenders	Spread	Adj. spread	Max. debt/EBITDA	Adj. max debt/EBITDA
Club	2.76** [2.33]	9.15 [0.40]	12.61 [0.51]	0.29 [0.62]	0.52 [1.04]
Ln (size)	1.43*** [3.05]	– 19.67*** [– 3.47]	– 24.18*** [– 3.48]	0.46*** [3.41]	– 0.02 [– 0.10]
EBITDA/assets	3.62 [0.71]	23.75 [0.20]	33.38 [0.25]	– 5.42 [– 1.48]	– 2.39 [– 0.64]
Post-LBO leverage	10.07*** [2.68]	46.24 [0.55]	41.83 [0.50]	2.98* [1.77]	2.90 [1.38]
Constant	2.76 [0.97]	208.82*** [3.29]	– 130.53** [– 2.01]	6.18*** [4.75]	– 1.88 [– 1.16]
Observations	85	83	83	43	43
R-squared	0.26	0.06	0.09	0.33	0.08

the deal. These data may provide an objective assessment of the financial credibility of offers made to target shareholders and explain differences in prices. We collect these data from Capital IQ and the SEC's Edgar database when available. Neither of these databases contains proxy statements for deals prior to 1994.

We present our findings from this analysis in Table 9. Panel A shows that the total amount of debt and equity commitments made to target shareholders is, on average, roughly equal to the total amount of funds needed to complete the deal—the average ratio of total commitments to total funds is 100.4%. This ratio is

somewhat higher in club deals (101.6%) than it is in sole-sponsored private equity deals (99.4%), but the difference is not statistically significant. Hence, there does not appear to be a large enough systematic difference in the financial credibility of offers as proxied by financing commitments made to target shareholders to explain their willingness to accept low prices in club deals. Panel A also reports the number of deals where total commitments fall below, exceed, or equal total funds (as presented in rows labeled *under*, *over*, and *equal*, respectively). Total commitments made are less than total funds needed in 15 out of 45 club deals (33%) compared to

19 out of 61 sole-sponsored private equity deals (31%). The minor difference in offer credibility captured by this measure is unlikely to explain low prices in club deals.

In Panel B, we present evidence on the mix of debt and equity commitments made to target shareholders. On average, equity commitments represent 35.6% of all commitments received. This ratio is higher in club deals (38.2%) than in sole-sponsored private equity deals (33.7%), but the difference is not statistically significant at conventional levels. These results are inconsistent with the hypothesis that clubs are able to benefit from higher leverage ratios as a result of attaching multiple private equity sponsors to a deal.

6.3. Size, risk, and debt terms

In this subsection, we provide more direct evidence on the view that club deals are motivated by capital constraints, diversification motives, or the ability to borrow at favorable terms.

Bearing in mind that our pricing results are robust to controls for target size, we investigate in more detail whether capital constraints appear to motivate club deals. While Table 4 shows that club deals are significantly larger on average than sole-sponsored LBOs, which is *prima facie* evidence that capital constraints may be a motivation for club deals, the difference is not statistically significant in the pre-2006 period (in which we find that club targets receive significantly lower premiums). In addition, we ask how many club deals are large enough that the private equity firms involved likely could not have acquired the target without pooling resources. Specifically, we ask how many club deals are such that the deal size (as measured by enterprise value (1) defined in Section 5) exceeds the largest sole-sponsored private equity deal done by the club members during a $[-2, +2]$ year window centered around the announcement date.

We find that only 16 of our 70 club deals (22.9%) meet this criterion, and the average premium in those deals (26.1%) is not statistically different from that of the other 54 deals (23.4%). Moreover, when we restrict the sample to club deals announced on or before December 31, 2005 (because 2006 and 2007 club deals lack the relevant two-year ahead comparison group), we find that only six of our 42 club deals (14.3%) meet the criterion, and the average premium in those deals (23.2%) is not statistically different from that of the other 36 deals (22.0%).

These facts cast some doubt on the notion that capital constraints are a primary motivation for the majority of club deals, especially before 2006, although it is possible that the degree of capital constraints faced by certain private equity firms may be dependent on deal characteristics that we cannot observe (but may be related to the debt financing terms that the sponsor(s) can obtain in loan markets).

Our evidence on target risk in Table 4 does not point to significant risk differences between targets in club and sole-sponsored private equity deals. Our two risk measures, return volatility and beta, do not differ significantly between the two categories of targets. As mentioned in our discussion of Table 4, neither does a measure of operating risk, historical cash flow volatility. We

further investigate whether differences in target complexity (as measured by the number of reported segments in Compustat segment files) or information asymmetry surrounding the target (as measured by analyst forecast errors in IBES) explain our findings. On average, club targets report 2.0 segments for the fiscal year prior to the deal announcement, which is statistically indistinguishable from 2.2 segments reported by sole-sponsored targets (despite the larger size of targets in club deals on average). We also find no significant differences in analyst forecast errors between the two types of targets.

To shed further light on these results on risk, and to explore whether certification to debt financiers is likely to be an important rationale for club formation, we next investigate whether debt financing terms are better or worse in club deals than they are in sole-sponsored private equity deals. Recent papers find that more reputable private equity firms (Demiroglu and James, 2010) and those with stronger bank relationships (Ivashina and Kovner, 2007) are able to obtain better terms on the debt financing for the LBOs they conduct. While Table 2 shows that a substantial fraction of our club deals is undertaken by the oldest and best-known private equity partnerships, a large fraction of our sole-sponsored deals is as well, so it is not clear *a priori* whether we should expect clubs to receive better debt financing terms.

If systematic differences in debt financing terms do exist, they could proxy for unobserved target characteristics. Specifically, if club deals have systematically worse debt financing terms than sole-sponsored private equity deals, then one may wonder whether unobserved target characteristics that drive up observed financing costs are also driving down target prices. If, on the other hand, club deals systematically have better debt financing terms than sole-sponsored private equity deals, this would be consistent with “benign” theories of club deals in which lenders offer superior terms when multiple private equity firms attach their names and reputations to a deal—though this would further deepen the puzzle about low prices paid to target shareholders in club deals.

To investigate these ideas, we link our sample of deals in SDC to credit facilities in DealScan using target and buyer names in SDC and all available name fields in DealScan.²⁴ Because we do this by electronic text matching, we further require that the facility start date in DealScan fall within a $[-30, +30]$ day window centered around the deal effective date in SDC. This procedure, along with a manual check to eliminate two false positives, results in 107 of our deals in SDC being matched to at least one credit facility in DealScan.

Panel C of Table 9 reports evidence on five measures, namely the number of lenders, spread over the London Interbank Offered Rate (LIBOR), adjusted spread (the spread minus the average spread for all LBO debt raised in the previous calendar quarter), the maximum debt-to-EBITDA allowed under the credit facility, and adjusted maximum debt-to-EBITDA (relative to LBO debt raised in the previous

²⁴ A caveat to this analysis is that DealScan may not identify all important sources of debt financing, such as mezzanine loans.

calendar quarter). On average, the number of lenders is higher in club deals (10.5) than in sole-sponsored private equity deals (7.5). The difference is statistically significant at the 5% level, and robust to controlling for transaction characteristics such as size, operating cash flow, and leverage in the transaction (reported in Panel D). The fact that club deals involve more lenders than sole-sponsored private equity deals is consistent with the allegation that clubs constrain the supply of debt financing for competing bids by aggressively locking up debt financiers. An alternative interpretation is that although we control for target size in the regression, the larger size of club deals results in more lenders.

The rest of our findings in Panels C and D suggest that club deals have insignificantly different spreads (raw and adjusted, with the univariate differences in Panel C and the regressions in Panel D pointing to a small difference in opposite directions) and insignificantly better maximum debt-to-EBITDA covenants (raw and adjusted) than sole-sponsored private equity deals do. Overall, the evidence is not strongly supportive of debt certification benefits in club deals, and does not support the notion that club targets are riskier in unobservable ways because we would expect greater risk to be reflected in worse financing terms.

7. Conclusion

In this paper, we show that acquisitions by clubs of prominent private equity acquirers are priced significantly lower than sole-sponsored private equity transactions and non-private equity merger and acquisition transactions. The differences are economically large: target shareholders receive approximately 10% less of pre-bid firm equity value, or roughly 40% lower premiums, in club deals compared to sole-sponsored LBOs. These results are robust to controls for observable target characteristics, including size, Q , measures of risk, and time and industry fixed effects. While our main results are based on target returns, we obtain similar results when we consider pricing measures based on acquisition multipliers.

We also find that the pricing differences are concentrated before 2006. Indeed, the club discount virtually disappears in 2006 and 2007. This result is potentially explained by the fact that the financial media began expressing concerns about club deals at the end of 2005 and the U.S. Department of Justice started an informal inquiry into the practice in 2006. We also find that the club deal discount is less pronounced in target firms with higher institutional ownership, suggesting that sophisticated institutional investors are able to bargain effectively with clubs. We find little support for motivations for club deals based on capital constraints, diversification motives, or the ability of clubs to obtain favorable debt amounts or prices. Of course, we are only able to investigate motivations for club formation that are related to observable target characteristics, and cannot rule in or out motivations based on target characteristics that are unobservable to the econometrician. In particular, the lower pricing of club deals may be an inadvertent byproduct of an unobserved benign motivation for club formation. In addition, because there does not exist a credible instrument in this setting, it is also possible that

there are unobservable factors that both drive club formation and can explain the level pricing result, the 2006 break, and the cross-sectional relation between club premiums and target institutional ownership, even in the absence of any reduction in competition.

We note that our sample consists exclusively of public-to-private transactions and that if one considered private-to-private transactions as well, the conclusions could be different. In fact, given the role of institutions in mitigating the club deal discount in public targets, we might not expect clubs to lead to low prices for private targets, which tend to have concentrated owners.

Our results on the pricing of club deals differ from those of Boone and Mulherin (2009) and Barger, Schlingemann, Stulz, and Zutter (2008). Boone and Mulherin (2009) examine the pricing of club deals in the 2003–2007 period, but, unlike us, find no evidence of a club deal discount in long-horizon acquisition premiums. Similarly, Barger, Schlingemann, Stulz, and Zutter (2008, footnote 4) report but do not tabulate that they find an insignificant club deal discount using long bid-period windows in a sample of 43 club deals conducted over 1980–2005. Neither of these studies distinguishes between prominent and other private equity firms, nor accounts for the important structural break in 2006 that we show (the Barger et al. sample ends in 2005). Broadening our focus to include transactions that SDC claims are LBOs but are not in our main sample of LBOs conducted by prominent private equity firms substantially reduces the magnitudes of the estimated club deal discounts both in the full 1984–2007 period and in the pre-2006 period, and renders the long-horizon premium difference statistically insignificant. Our results combine with those of these authors to suggest that the pre-2006 club deal discount we report is likely specific to large, prominent private equity firms.

Finally, our results on pricing in club deals do not rule out the possibility that the transactions are economically efficient even if there is room for target shareholders to share more in the surplus. Club deals may, in fact, be socially beneficial if their pooling of multiple expert opinions helps redeploy target assets to more productive uses. Several studies of the economic impact of private equity show substantial long-term benefits to society (Gurung and Lerner, 2008). Whether club deals are different from sole-sponsored LBOs along this dimension is an important question for future research.

Appendix. List of private equity deals

This table lists the 201 deals conducted by prominent private equity firms in our sample. Panel A lists club deals and Panel B lists sole-sponsor deals. We list a maximum of two private equity acquirers, with prominent private equity firms listed first (in order of appearance in Panel A of Table 2). If there are more than two private equity acquirers, the others are denoted “et al.” In the Table A1, WCAS abbreviates Welsh, Carson, Anderson, and Stowe.

Table A1

Year	Target name	Private equity acquirers	Year	Target name	Private equity acquirers
<i>Panel A: Club deals</i>					
1985	SCOA Industries Inc.	Thomas H. Lee Partners, Drexel Burnham Lambert	2005	Neiman Marcus Group Inc.	TPG, Warburg Pincus
1986	PT Components Inc.	Merrill Lynch, Madison Dearborn, et al.	2005	Party City Corp	Berkshire Partners, Weston Presidio Capital
1986	Revco DS Inc.	JP Morgan, Salomon Brothers	2005	ShopKo Stores Inc.	Sun Capital Partners, Lubert-Adler
1987	Decision Industries Corp	WCAS, JH Whitney	2005	SunGard Data Systems Inc.	KKR, Blackstone, et al.
1988	Wickes Cos Inc.	Blackstone, Wasserstein Perella	2005	Town & Country Trust	Morgan Stanley, Onex
1989	CNW Corp	Blackstone, DLJ	2005	Toys "R" Us Inc.	KKR, Bain Capital
1993	Digital Communications Assoc	WCAS, Sprout Group	2005	UICI	Blackstone, Goldman Sachs, et al.
1995	Riverwood International Corp	Clayton, Dubilier & Rice, 1818 Fund, et al.	2006	ADESA Inc.	Goldman Sachs, Partheon Capital, et al.
1995	Rockefeller Center Properties	Goldman Sachs, Exor Group, et al.	2006	Albertsons Inc.	Cerberus Capital Management, Lubert-Adler
1996	Leslie's Poolmart Inc.	Leonard Green & Partners, Hancock Park Associates	2006	ARAMARK Corp	Goldman Sachs, Thomas H. Lee Partners, et al.
1997	Fisher Scientific Intl Inc.	Merrill Lynch, Thomas H. Lee Partners, et al.	2006	Biomet Inc.	KKR, Blackstone, et al.
1997	Multicare Cos Inc.	TPG, Cypress Group	2006	Direct General Corp	TPG, Fremont Partners
1997	Telemundo Group Inc.	Apollo Management, Bastion Capital	2006	Education Management Corp	Goldman Sachs, Providence Equity Partners
1998	Centennial Cellular Corp	Blackstone, WCAS	2006	Freescall Semiconductor Inc.	Blackstone, TPG, et al.
1998	CompDent Corp	GTCR Golder Rauner, TA Associates, et al.	2006	HCA Inc.	KKR, Merrill Lynch, et al.
1998	MedCath Inc.	KKR, WCAS	2006	Intergraph Corp	TPG, Hellman & Friedman, et al.
1998	Regal Cinemas Inc.	KKR, HM Capital Partners	2006	Kinder Morgan Inc.	Goldman Sachs, The Carlyle Group
1998	Republic Engineered Steels	Blackstone, Veritas Capital Partners	2006	Michaels Stores Inc.	Blackstone, Bain Capital
1999	Berkshire Realty Co Inc.	Blackstone, Whitehall Street (Goldman Sachs)	2006	Open Solutions Inc.	The Carlyle Group, Providence Equity Partners
1999	Big Flower Holdings Inc.	Thomas H. Lee Partners, Evercore Capital Partners	2006	OSI Restaurant Partners Inc.	Bain Capital, Catterton Partners
1999	Genesis Health Ventures Inc.	TPG, Cypress Group	2006	Petco Animal Supplies Inc.	TPG, Leonard Green & Partners
1999	Integrated Circuit Systems Inc.	Bain Capital, Bear Stearns	2006	Readers Digest Association Inc.	Merrill Lynch, GSO Capital Partners
1999	Wilmar Industries Inc.	JP Morgan (Chase Capital), Sterling Inv. Partners	2006	Sabre Holdings Corp	TPG, Silver Lake Partners
2000	Brookdale Living	Fortress Investment Group, Capital Z Partners	2006	Trizec Properties Inc.	Blackstone, Brookfield Asset Management
2000	Petco Animal Supplies Inc.	TPG, Leonard Green & Partners	2006	Univision Communications Inc.	TPG, Thomas H. Lee Partners, et al.
2002	Exco Resources	Cerberus, Greenhill Capital Partners	2006	West Corp	Thomas H. Lee Partners, Quadrangle Group
2002	National Golf Properties Inc.	Goldman Sachs, Starwood Capital Partners	2007	Alltel Corp	TPG, Goldman Sachs
2002	Quintiles Transnational Corp	TPG, One Equity Partners	2007	Avaya Inc.	TPG, Silver Lake Partners
2004	AMC Entertainment Inc.	Apollo Management, JP Morgan	2007	Dollar General Corp	KKR, Goldman Sachs
2004	Metro-Goldwyn-Mayer Inc.	TPG, Providence Equity Partners, et al.	2007	Eagle Hosp Prop Trust Inc.	Apollo Management, JF Capital Advisors
2004	PanAmSat Corp	KKR, The Carlyle Group, et al.	2007	Kronos Inc.	Hellman & Friedman, JMI Equity
2004	Select Medical Corp	WCAS, GTCR Golder Rauner	2007	Laureate Education Inc.	KKR, Citigroup Private Equity, et al.
2004	Texas Genco Holdings Inc.	KKR, Blackstone, et al.	2007	Nuveen Investments Inc.	Madison Dearborn Partners, Citigroup
2005	DoubleClick Inc.	Hellman & Friedman, JMI Equity	2007	Topps Co Inc.	Madison Dearborn Partners, Tornante Co.
2005	Linens n Things Inc.	Apollo Management, Silver Point Capital, et al.	2007	TXU Corp	KKR, TPG, et al.
<i>Panel B: Sole-sponsor deals</i>					
1984	Denny's Inc.	Merrill Lynch	1994	Borden Inc.	KKR
1984	Palm Beach Inc.	Merrill Lynch	1995	Bruno's Inc.	KKR
1985	Beatrice Companies Inc.	KKR	1995	Diagnostek Inc.	Warburg Pincus

Table A1 (continued)

Panel B: Sole-sponsor deals					
1985	Jack Eckerd Corp	Merrill Lynch	1996	Belden & Blake Corp	TPG
1985	National Gypsum Co	Goldman Sachs	1996	Community Health Systems Inc.	Forstmann Little
1985	Storer Communications Inc.	KKR	1996	Kinder-Care Learning Centers	KKR
1985	UNIROYAL Inc.	Clayton, Dubilier & Rice	1996	Syratech Corp	Thomas H. Lee Partners
1986	Fruehauf Corp	Merrill Lynch	1996	Triad Systems Corp	HM Capital Partners
1986	Lear Siegler Inc.	Forstmann Little	1997	Amscan Holdings Inc.	Goldman Sachs
1986	Midland-Ross Corp	Forstmann Little	1997	CommNet Cellular Inc.	Blackstone
1986	Owens-Illinois Inc.	KKR	1997	Control Data Systems Inc.	WCAS
1986	Pandick Inc.	Morgan Stanley	1997	Dynatech Corp	Clayton, Dubilier & Rice
1986	Safeway Stores Inc.	KKR	1997	ERO Inc.	HM Capital Partners
1986	Sheller Globe Corp	Lehman Brothers Private Equity	1997	Hechinger Co	Leonard Green & Partners
1986	Sybron Corp	Forstmann Little	1997	LIN Television Corp	HM Capital Partners
1987	Becor Western Inc.	Goldman Sachs	1997	Melamine Chemicals Inc.	KKR
1987	Borg-Warner Corp	Merrill Lynch	1997	Smith's Food & Drug Centers	KKR
1987	Burlington Industries Inc.	Morgan Stanley	1997	Tuesday Morning Corp	Madison Dearborn Partners
1987	Comdata Network Inc.	WCAS	1997	Zilog Inc.	TPG
1987	Jim Walter Corp	KKR	1998	MTL Inc.	Apollo Management
1987	Seaman Furniture Co	KKR	1998	Rival Co	Berkshire Partners
1987	Supermarkets General Corp	Merrill Lynch	1998	SFX Broadcasting Inc.	HM Capital Partners
1988	American Health Cos Inc.	Thomas H. Lee Partners	1999	BancTec Inc.	WCAS
1988	Colt Industries Inc.	Morgan Stanley	1999	Blount International Inc.	Lehman Brothers Private Equity
1988	Cullum Cos Inc.	Morgan Stanley	1999	Concentra Managed Care Inc.	WCAS
1988	Fort Howard Corp	Morgan Stanley	1999	Empi Inc.	The Carlyle Group
1988	Hyponex Corp	Clayton, Dubilier & Rice	1999	Physicians Specialty Corp	TA Associates
1988	Koppers Co Inc.	Lehman Brothers Private Equity	1999	Walden Residential Properties	HM Capital Partners
1988	Pullman Co	Forstmann Little	1999	White Cap Industries Inc.	Leonard Green & Partners
1988	RJR Nabisco Inc.	KKR	2000	Coinmach Laundry Corp	GTCR Golder Rauner
1988	Stanadyne Inc.	Forstmann Little	2000	Mark IV Industries Inc.	BC Partners
1988	Stop & Shop Cos Inc.	KKR	2000	Veterinary Centers of America	Leonard Green & Partners
1989	General Nutrition Inc.	Thomas H. Lee Partners	2001	Citadel Communications Corp	Forstmann Little
1989	Jefferson Smurfit Corp	Morgan Stanley	2002	AmeriPath Inc.	WCAS
1990	Aristech Chemical Corp	Blackstone	2002	Ebenx Inc.	WCAS
1990	General Instrument Corp	Forstmann Little	2003	FTD Inc.	Leonard Green & Partners
1990	Healthco International Inc.	HM Capital Partners	2003	Varsity Brands Inc.	Leonard Green & Partners
1993	Alden Press Co	KKR	2004	Boca Resorts Inc.	Blackstone
1993	Preferred Health Care Ltd	Warburg Pincus	2004	Extended Stay America Inc.	Blackstone
2004	LNR Property Corp	Cerberus Capital Management	2006	SITEL Corp	Onex
2004	Prime Hospitality Corp	Blackstone	2006	SourceCorp Inc.	Apollo Management
2004	Sola International Inc.	EQT Partners	2006	Sports Authority Inc.	Leonard Green & Partners
2004	US Oncology Inc.	WCAS	2006	TransMontaigne Inc.	Morgan Stanley
2005	Amlti Residential Ppty Trust	Morgan Stanley	2006	Water Pik Technologies Inc.	The Carlyle Group
2005	Insight Communications Co	The Carlyle Group	2006	Yankee Candle Co Inc.	Madison Dearborn Partners
2005	Instinet Group Inc.	Silver Lake Partners	2007	Archstone-Smith Trust	Lehman Brothers Private Equity
2005	La Quinta Corp	Blackstone	2007	Bausch & Lomb Inc.	Warburg Pincus
2005	Metals USA Inc.	Apollo Management	2007	Catalina Marketing Corp	Hellman & Friedman
2005	SERENA Software Inc.	Silver Lake Partners	2007	CDW Corp	Madison Dearborn Partners
2005	Specialty Laboratories Inc.	WCAS	2007	Ceridian Corp	Thomas H. Lee Partners
2005	SS&C Technologies Inc.	The Carlyle Group	2007	Claire's Stores Inc.	Apollo Management
2005	Worldwide Rest Concepts Inc.	Pacific Equity Partners	2007	Crescent Real Estate Equities	Morgan Stanley
2005	Wyndham International Inc.	Blackstone	2007	EGL Inc.	Apollo Management
2006	Aleris International Inc.	TPG	2007	Equity Inns Inc.	Goldman Sachs
2006	Burlington Coat Factory	Bain Capital	2007	First Data Corp	KKR
2006	CarrAmerica Realty Corp	Blackstone	2007	Florida East Coast Inds Inc.	Fortress Investment Group
2006	Columbia Equity Trust Inc.	JP Morgan	2007	Guitar Center Inc.	Bain Capital
2006	Encore Medical Corp	Blackstone	2007	Hilton Hotels Corp	Blackstone
2006	Equity Office Properties Trust	Blackstone	2007	Innkeepers USA Trust	Apollo Management
2006	Glenborough Realty Trust Inc.	Morgan Stanley	2007	Interpool Inc.	Fortress Investment Group
2006	Jacuzzi Brands Inc.	Apollo Management	2007	ServiceMaster Co	Clayton, Dubilier & Rice
2006	Laserscope Inc.	Warburg Pincus	2007	Smart & Final Inc.	Apollo Management
2006	Marsh Supermarkets	Sun Capital Partners	2007	Synagro Technologies Inc.	The Carlyle Group
2006	MeriStar Hospitality Corp	Blackstone	2007	United Surgical Partners Intl	WCAS
2006	RailAmerica Inc.	Fortress Investment Group	2007	USI Holdings Corp	Goldman Sachs
2006	Realogy Corp	Apollo Management			

References

- Abadie, A., Imbens, G., 2007. Bias corrected matching estimators for average treatment effects. Unpublished Working Paper, Harvard University.
- Axelsson, U., Jenkinson, T., Strömberg, P., Weisbach, M., 2007. Leverage and pricing in buyouts: an empirical analysis. Unpublished Working Paper, Swedish Institute for Financial Research.
- Bargeron, L., Schlingemann, F., Stulz, R., Zutter, C., 2008. Why do private acquirers pay so little compared to public acquirers? *Journal of Financial Economics* 89, 375–390.
- Bailey, E., 2007. Are private equity consortia anticompetitive? The economics of club bidding. Antitrust Source.
- Boone, A., Mulherin, H., 2009. Do private equity consortiums facilitate collusion in takeover bidding? Unpublished Working Paper, University of Kansas.
- Brander, J., Amit, R., Antweiler, W., 2002. Venture capital syndication: improved venture selection vs. the value-added hypothesis. *Journal of Economics and Management Strategy* 11, 423–452.
- Comment, R., Schwert, G., 1995. Poison or placebo? evidence on the deterrence and wealth effects of modern antitakeover measures. *Journal of Financial Economics* 39, 3–43.
- Cramton, P., Schwartz, J., 2000. Collusive bidding: lessons from the FCC spectrum auctions. *Journal of Regulatory Economics* 17, 229–252.
- DeAngelo, L., 1986. Accounting numbers as market valuation substitutes: a study of management buyouts of public stockholders. *The Accounting Review* 61, 400–420.
- DeAngelo, H., DeAngelo, L., Rice, E., 1984. Going private: minority freezeouts and stockholder wealth. *Journal of Law and Economics* 27, 367–401.
- Demiroglu, C., James, C., 2010. The role of private equity group reputation in LBO financing. *Journal of Financial Economics* 96, 306–330.
- Du, Q., 2008. Birds of a feather or celebrating differences? the formation and impact of venture capital syndication. Unpublished Working Paper, University of British Columbia.
- Gaspar, J., Massa, M., Matos, P., 2005. Shareholder investment horizons and the market for corporate control. *Journal of Financial Economics* 76, 135–165.
- Gompers, P., Ishii, J., Metrick, A., 2003. Corporate governance and equity prices. *Quarterly Journal of Economics* 118, 107–155.
- Gompers, P., Lerner, J., 2000. Money chasing deals? The impact of fund inflows on the valuation of private equity investments. *Journal of Financial Economics* 55, 281–325.
- Graham, D., Marshall, R., 1989. Collusive bidder behavior at single-object second-price and english auctions. *Journal of Political Economy* 95, 1217–1239.
- Guo, S., Hotchkiss, E., Song, W., 2009. Do buyouts (still) create value? *Journal of Finance*, forthcoming.
- Gurung, A., Lerner, J., 2008. The global economic impact of private equity. World Economic Forum.
- Hendricks, K., Porter, R., 1992. Joint bidding in federal OCS auctions. *American Economic Review* 82, 506–511.
- Hochberg, Y., Ljungqvist, A., Lu, Y., 2009. Networking as a barrier to entry and the competitive supply of venture capital. *Journal of Finance* 65, 829–859.
- Ivashina, V., Kovner, A., 2007. The private equity advantage: private equity firms and relationship banking. Unpublished Working Paper, Harvard University.
- Jensen, M., 1986. Agency costs of free cash flow, corporate finance, and takeovers. *American Economic Review* 76, 323–329.
- Kaplan, S., 1989a. The effects of management buyouts on operating performance and value. *Journal of Financial Economics* 24, 217–254.
- Kaplan, S., 1989b. Management buyouts: evidence on taxes as a source of value. *Journal of Finance* 44, 611–632.
- Kaplan, S., Ruback, R., 1995. The valuation of cash flow forecasts: an empirical analysis. *Journal of Finance* 50, 1059–1093.
- Kaplan, S., Stein, J., 1993. The evolution of buyout pricing and financial structure in the 1980s. *Quarterly Journal of Economics* 108, 313–357.
- Kaplan, S., Strömberg, P., 2009. Leveraged buyouts and private equity. *Journal of Economic Perspectives* 23, 121–146.
- Kaplan, S., Zingales, L., 1997. Do investment-cash flow sensitivities provide useful measures of financing constraints? *Quarterly Journal of Economics* 112, 169–215.
- Lang, L., Stulz, R., Walking, R., 1989. Managerial performance, Tobin's Q, and the gains from successful tender offers. *Journal of Financial Economics* 24, 137–154.
- Lerner, J., 1994. The syndication of venture capital investments. *Financial Management* 23, 16–27.
- Marquez, R., Singh, R., 2009. The economics of club bidding in private equity. Unpublished Working Paper, Boston University.
- Officer, M., 2003. Termination fees in mergers and acquisitions. *Journal of Financial Economics* 69, 431–467.
- Officer, M., 2007. The price of corporate liquidity: acquisition discounts for unlisted targets. *Journal of Financial Economics* 83, 571–598.
- Perry, S., Williams, T., 1994. Earnings management preceding management buyout offers. *Journal of Accounting and Economics* 18, 157–179.
- Phalippou, L., 2009. Beware of venturing into private equity. *Journal of Economic Perspectives* 23, 147–166.
- Ritter, J., 2008. Forensic finance. *Journal of Economic Perspectives* 22, 127–147.
- Schwert, G., 1996. Markup pricing in mergers and acquisitions. *Journal of Financial Economics* 41, 153–219.